

IDEA for Teachers: Tutorial

Kolibri, Nextcloud, Wikipedia offline, Files Disk and Backup Disk

Version 2.1, 13 November 2026

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1. What IDEA is

IDEA (Initiative for Digital Education in Africa) brings offline computing to rural schools. A small computer in the school, the **IDEA box**, holds ready-made lessons in Kolibri, a place for your own files in Nextcloud, and the whole of Wikipedia. Teachers and learners open them from any classroom computer, laptop or phone on the school Wi-Fi, without internet. This tutorial shows you, step by step, how to use them in your lessons.

> **No internet needed.** Everything in this guide works inside the school.

1.1 Words used in this guide

Word	Meaning
IDEA box	The small Raspberry Pi computer in the school that runs the apps. Each box has its own name, on its sticker. The kit's three boxes are **sunny-robin**, **happy-zebra** and **kind-panda**.
Console	The web page that lists the apps and opens them.
Demo apps	**Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: practice copies with example content, on every IDEA box. Use them to follow this tutorial (1.3).
IDEA shortcut	The one shortcut on a classroom computer's desktop. It finds the IDEA box and opens the Console (chapter 2).
System Disk	The IDEA box's own disk. The apps live on it.
IDEA Disk	A USB disk or stick that the Console has prepared. It can become a **Files Disk** or a **Backup Disk**.
Spare disk	An IDEA Disk that is still empty.
Files Disk	An IDEA Disk for your own files. It appears as the folder **School Files** in Nextcloud (chapter 8).
Backup Disk	An IDEA Disk that keeps copies of the apps' data (chapter 9).
Operator	The person at the school who looks after the IDEA box (IDEA for Operators: Tutorial).

1.2 The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (on the kit, for example CK-12, PhET, OpenStax and African Storybook). You make classes, lessons and quizzes and see each learner's progress.	3, 4
Nextcloud	Your own files: share worksheets with a class, write documents together in **Nextcloud Office**, collect homework.	5, 6, 8

App	What you use it for	Chapters
Wikipedia	The encyclopedia, offline: the full English Wikipedia, and a Simple English one in the Demo.	7

1.3 How to use this guide

- This tutorial is for teachers. The school operator has its own: IDEA for Operators: Tutorial.
- Each chapter has numbered steps with screenshots. The screenshots use the demo setup that is on every IDEA box: the demo school **IDEA Demo School** with class **Grade 5** and the demo accounts in the appendix (`admin`, `teacher1`, `teacher2`, `learner1` to `learner4`; teachers and the admin have a password: the account name followed by `911!`; learners sign in in Kolibri with only their username). Your school's names will be different.
- **Practise in the Demo apps.** Every IDEA box of the kit has six apps: the school's own **Kolibri**, **Nextcloud** and **Wikipedia**, and a Demo copy of each. Follow this tutorial in **Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: they hold the demo school, the demo accounts and the example content. The school's own apps stay clean for real lessons.

1.4 How IDEA looks after the kit

This section explains what IDEA counts about the kit, and why.

> IDEA counts how the kit is used, not who uses it or what they write, so that it can offer help. A donated kit may move to another school if it stays unused.

2. Getting connected and opening an app

All apps open from one web page: the **Console**. How you get to the Console depends on the computer you use.

2.1 Connect to the Wi-Fi

The classroom computers are already connected. Do this only on your own laptop, tablet or phone.

The IDEA router at your school makes two Wi-Fi networks. Their names and first passwords are the same at every IDEA school:

Network	Password (until your school changes it)	Who joins it	What it reaches
IDEA	`idea911!`	Students, and the classroom computers	Only the IDEA box: the Console and its apps. No internet.
IDEA-Teachers	`idea-teachers911!`	Teachers only	The IDEA box, and the internet. Internet is **on**; your IDEA contact can switch it off when the head teacher asks.

- Give students only the **IDEA** password. Keep the **IDEA-Teachers** password among the teachers.
- The classroom computers join **IDEA** by themselves. To put a teacher's classroom computer on **IDEA-Teachers**, choose it in the computer's Wi-Fi menu and type its password.

To connect:

1. Open the Wi-Fi settings on your device.
2. Choose **IDEA-Teachers** (teachers) or **IDEA** (students).
3. Type the password and join.

2.2 Open the Console

There are two ways to open the Console. Use the one that fits the device in front of you.

2.2.1 On a classroom computer

The classroom computers have **one** shortcut on the desktop, named **IDEA**. There are no separate icons for Kolibri, Nextcloud or the other apps.

1. Double-click **IDEA** on the desktop. You can also find it in the app menu. It opens in Chrome.

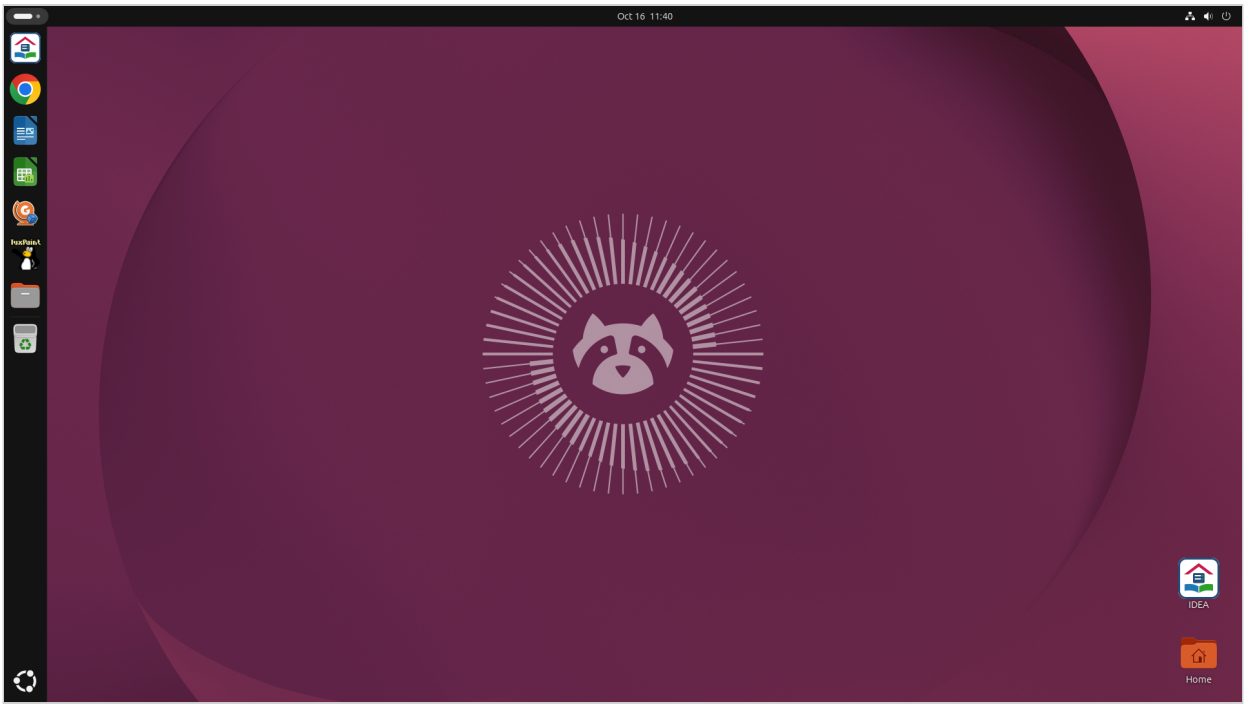


Figure 1. The classroom computer desktop with the one IDEA shortcut

2. A start page titled **IDEA** appears for a moment and says **Looking for the school server…**.

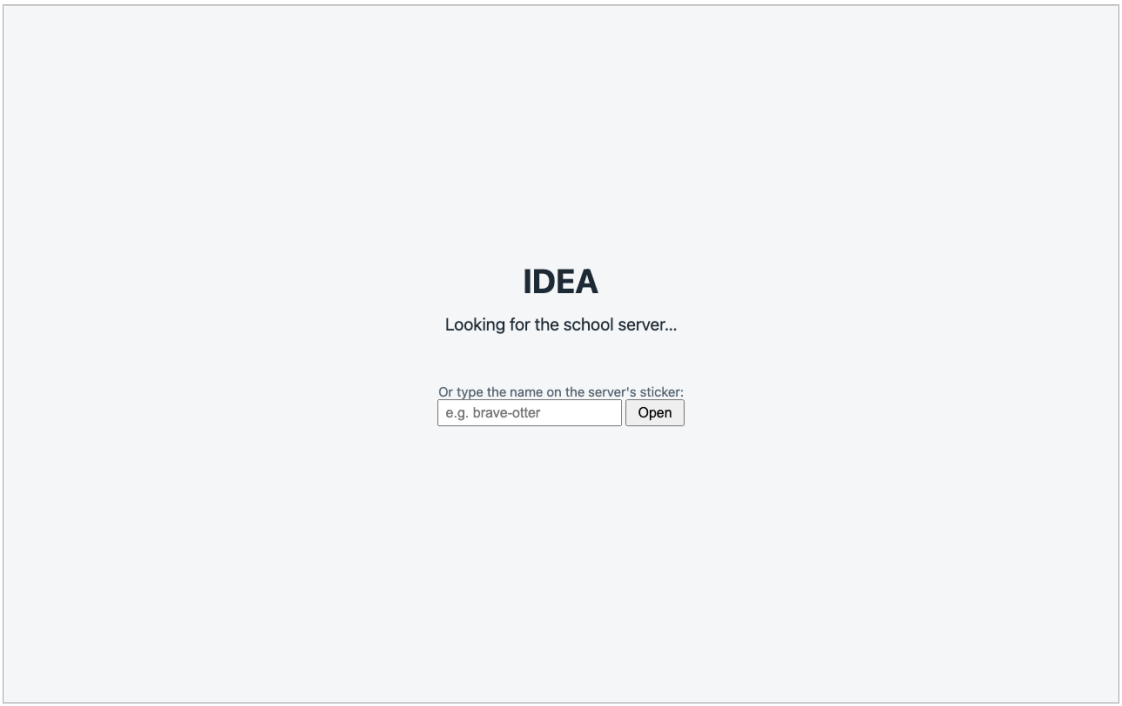


Figure 2. The IDEA start page: Looking for the school server…

3. The Console opens by itself.

If the start page says **No school server was found**, check that the computer is connected to the school network. Then type the first line of the box's sticker, for example `brave-otter`, in the field **Or type the name on the server's sticker**. Use small letters, with `-` between the words. Press **Open**. The start page remembers that IDEA box.

2.2.2 On your own laptop, tablet or phone

1. Connect to the Wi-Fi (2.1).
2. Open the browser.
3. Type `http://idea.local` in the address bar and press Enter.
4. The Console opens.

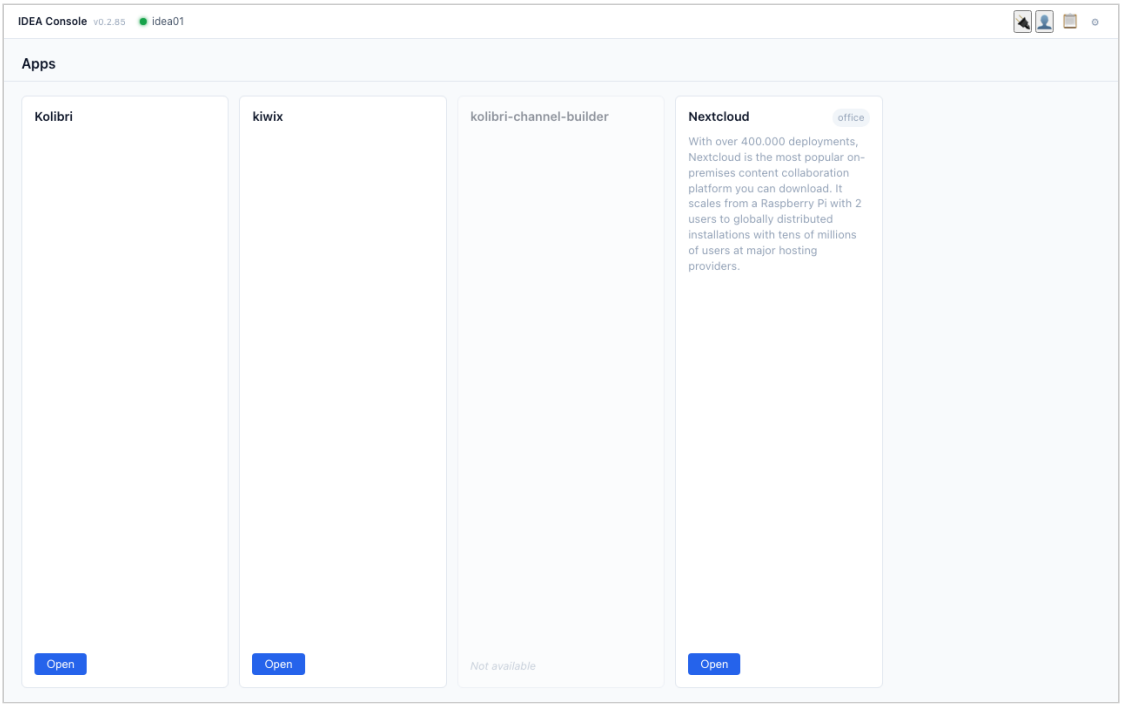


Figure 3. The Console in the browser

5. Bookmark it, so you can open it again with one click. In Chrome, press **Ctrl+D** (on a Mac ⌘+**D**), then **Done**.

If the Console does not open, type the address on the second line of the box's sticker instead, for example `http://brave-otter.local`.

> **Two IDEA boxes at your school?** The Console shows the apps of all of them. Each card says on which box the app runs, for example **on kind-panda**.

2.3 First steps in the Console

1. The Console opens on the **Apps** page: one card per app, each with a short description and a blue **Open** button. The green dot at the top shows the Console is connected to the IDEA box.

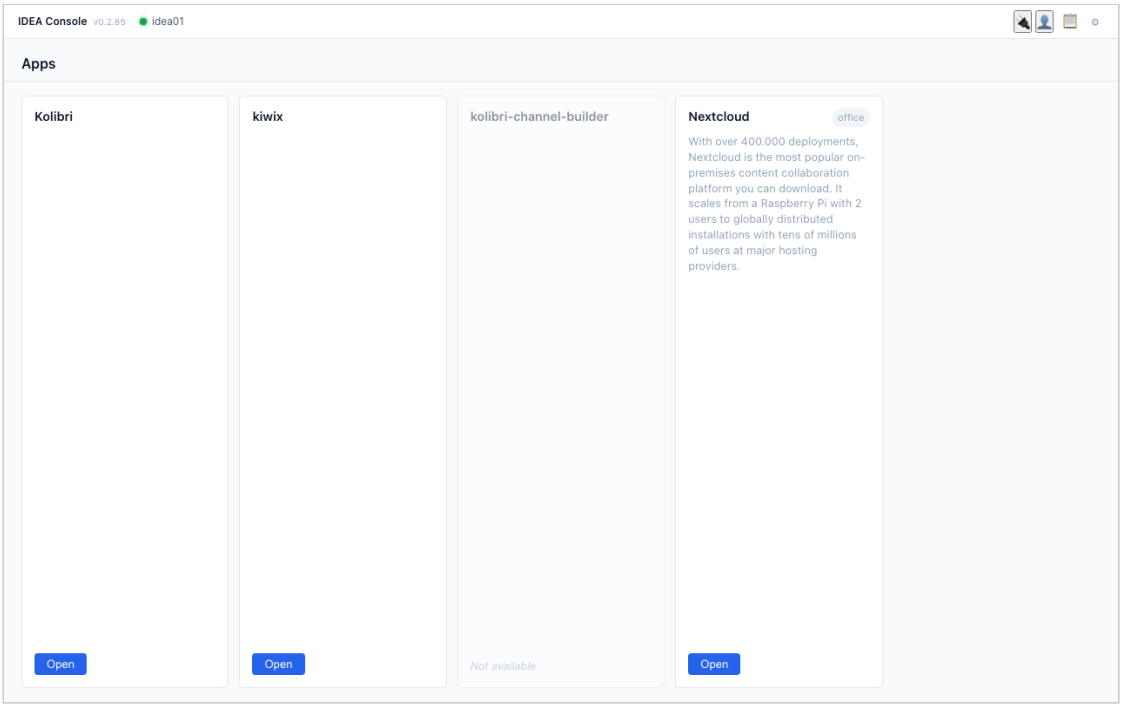


Figure 4. Console Apps page (two of the cards)

2. Click the blue **Open** button on the card of the app you need. The app opens in a new tab. Here it is Kolibri's sign-in page.

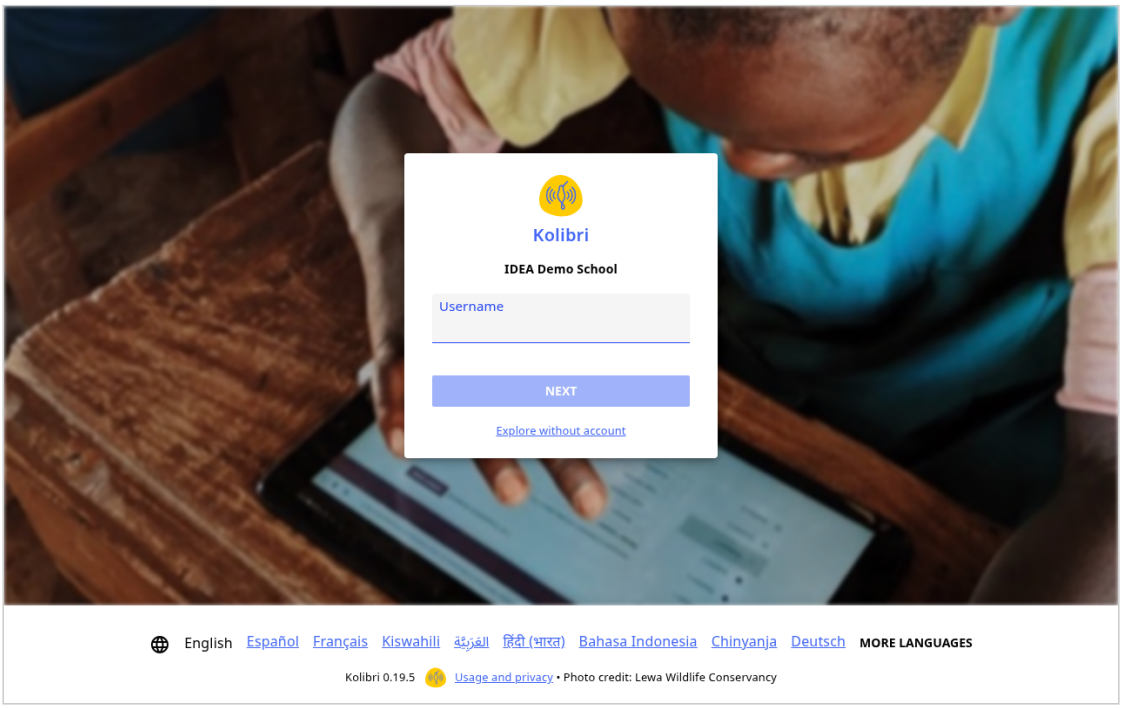


Figure 5. Kolibri opened from the Console

Good to know:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your school operator.
- **Known problem: apps that don't open.** If an IDEA box is switched off, or was once connected to yours and is now at another school, the Console still shows its apps as if they worked. **Open** then gives a page that doesn't load. There is no fix yet. Use the apps that are **on** your own school's IDEA box (the name on its sticker).
- **The classroom computers reset at every restart.** Anything saved on the computer itself is gone after a restart. Save work in Nextcloud (chapter 6).
- Do not type numbers (addresses or ports) into the address bar. Always start from the Console and use **Open**.
- A student computer that no longer works properly is not repaired: the operator resets its card (IDEA for Operators: Tutorial). Students lose only what they saved on the computer itself, so let them save their work in Nextcloud (chapter 6).
- The person icon at the top right (**Operator Login**) is for the operator (IDEA for Operators: Tutorial). Teachers do not need it.

> **Tip.** You can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

About the examples in this chapter. Every IDEA box comes with the demo setup switched on: the app **Kolibri Demo**, with the school **IDEA Demo School**, the class **Grade 5** and the demo accounts. The screens in this chapter come from it. Open the **Kolibri Demo** card in the Console, not the **Kolibri** card. The steps work the same in the school's own Kolibri, with your own school, class and account names. The demo accounts are listed in Appendix: demo accounts at the end of this tutorial.

In the demo, you sign in as `teacher1` (or `teacher2`). The Library has the channels **Maths**, **Grade 5 Science** and **Early Reading**, and also a **Khan Academy (English - US curriculum)** channel if you want more practice later. This chapter uses **Maths**. The ready-made **Maths lesson** for Grade 5 includes the Fractions items (pictures, short videos and **Fractions practice**) and the rest of the Maths channel. Section 3.4 shows how you would make a smaller Fractions lesson of your own from **Maths > Fractions**. The demo quiz is **Grade 5 Practice Quiz**.

3.1 Signing in

This is how you sign in to Kolibri Demo as a teacher, so you can try the steps that follow.

Open **Kolibri Demo** from the Console (2.3). Type your username and click **NEXT**, then type your password and click **Sign in**.

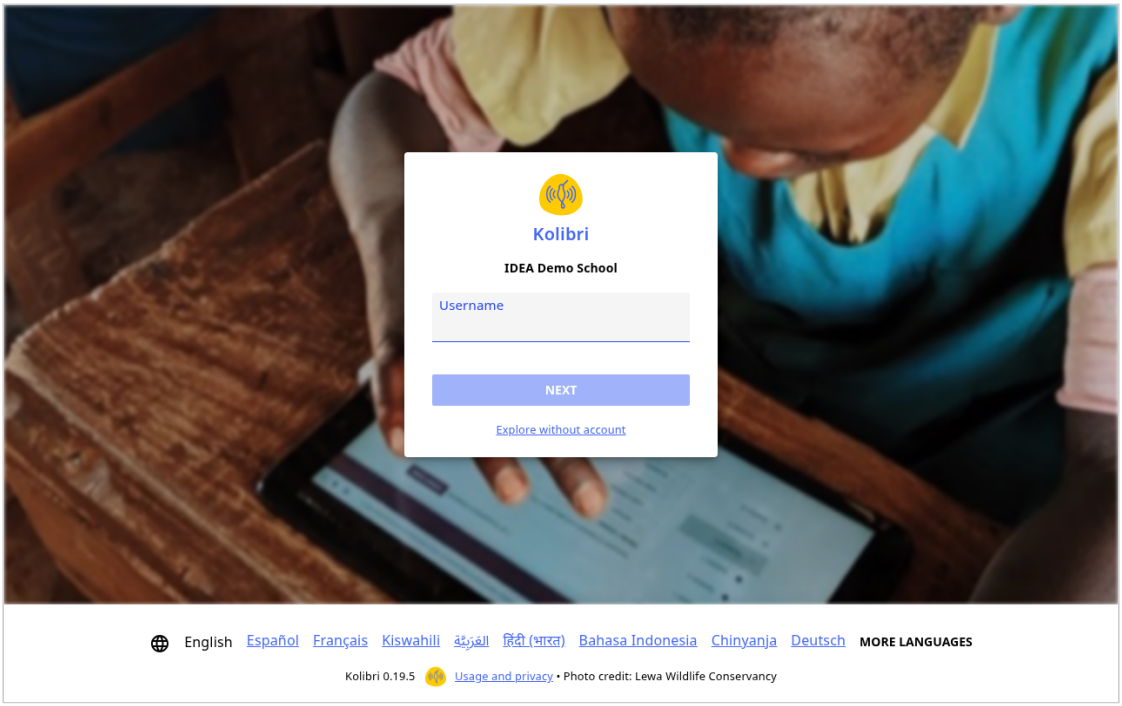


Figure 6. Sign in to Kolibri Demo

3.2 Create a class

A class is the group you teach in Kolibri. This section shows you how to make one.

Creating classes needs a Kolibri admin account (in the demo school, `admin`, password `admin911!`; `teacher1` is the coach of Grade 5).

1. Open **Facility > Classes**. You see the list of classes.

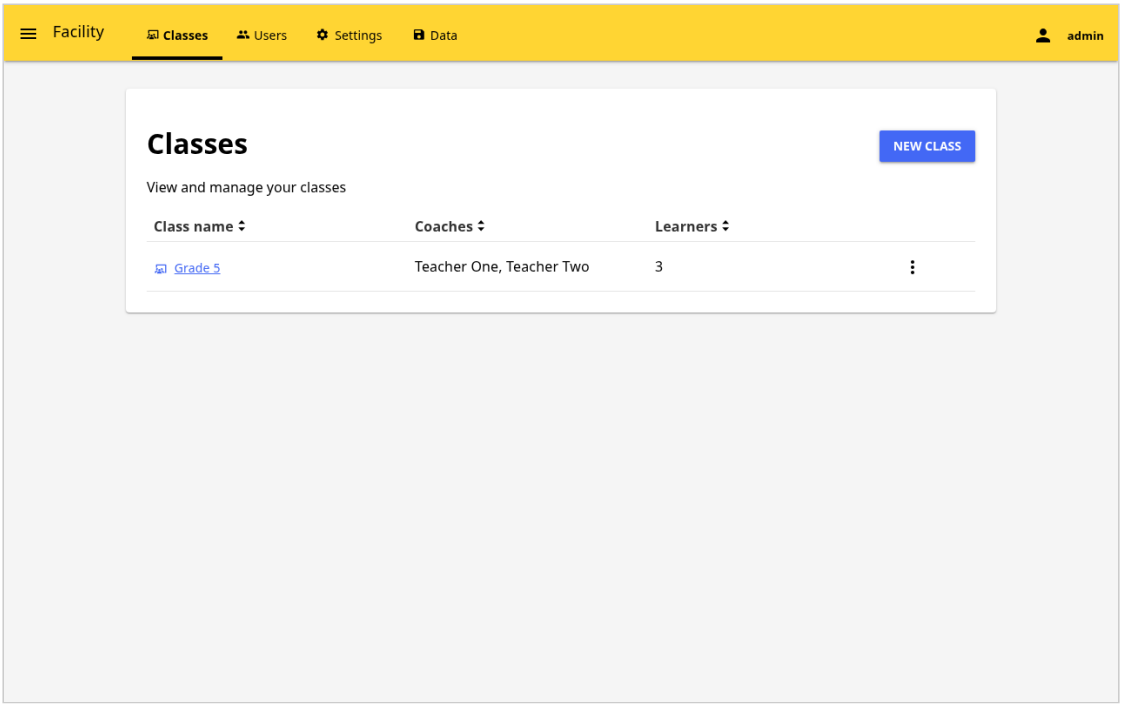


Figure 7. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed **CANCEL** because `Grade 5` already exists.)

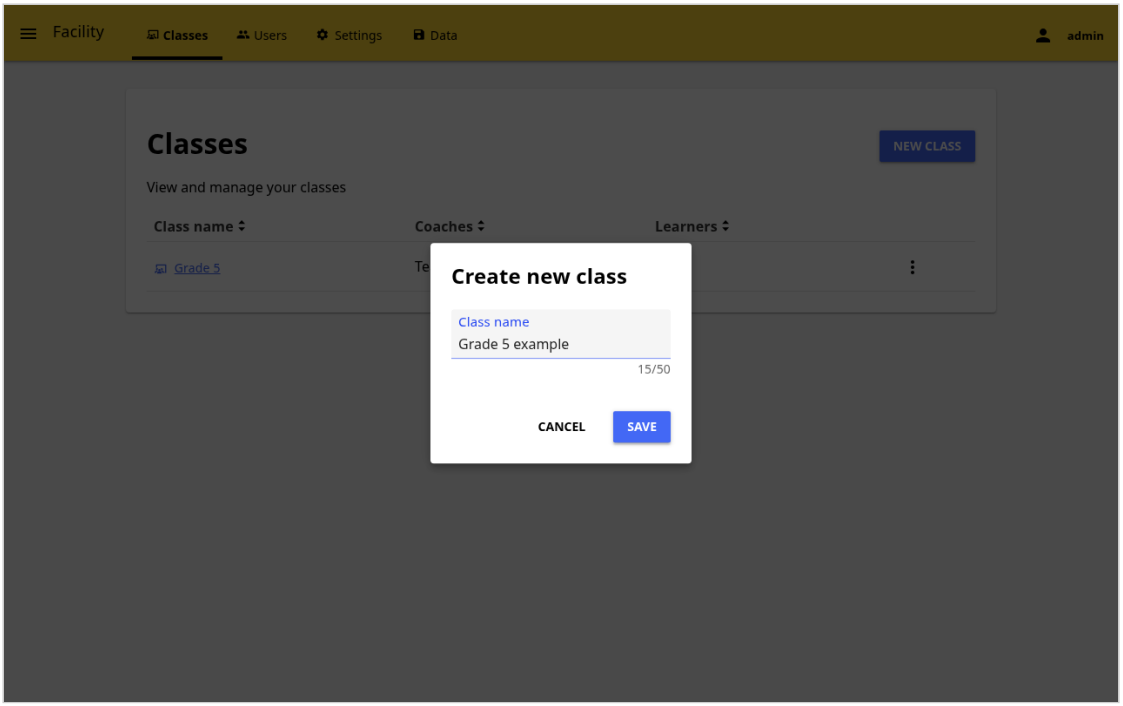


Figure 8. Create new class

3.3 Enroll learners

This is how you add learners to a class.

1. Click the class name. On the class page (with **Coaches** and **Learners**), click **ENROLL LEARNERS**. The enroll page opens. It only lists people who are not yet in the class (in the Demo that includes **Learner Four**, and it may also list the admin account — tick only the learner you want).

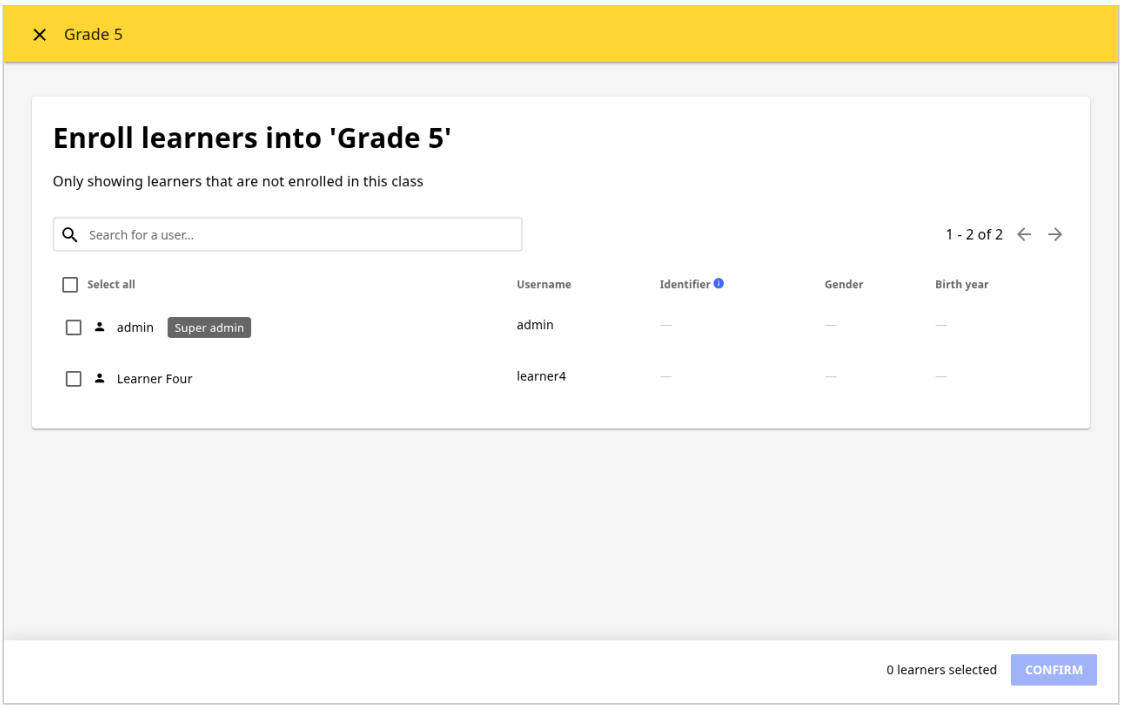


Figure 9. Enroll learners page

2. Tick each learner you want to add (here: **Learner Four**), then click **CONFIRM**. (In the demo we closed the page without confirming, so Grade 5 stays as it is.)

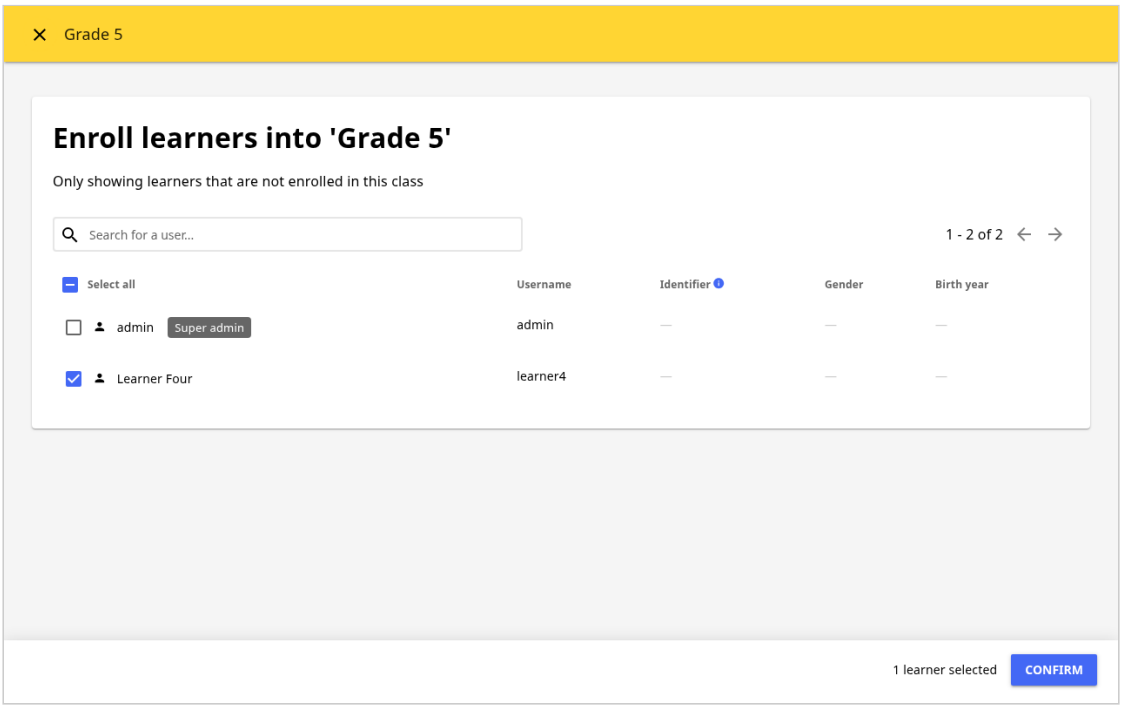


Figure 10. Learner ticked

3. Learners already in the class appear under **Learners**. In the Demo, Grade 5 already has Learner One, Learner Two and Learner Three. Learner Four is in the school but not in this class, so you can practise the step above.

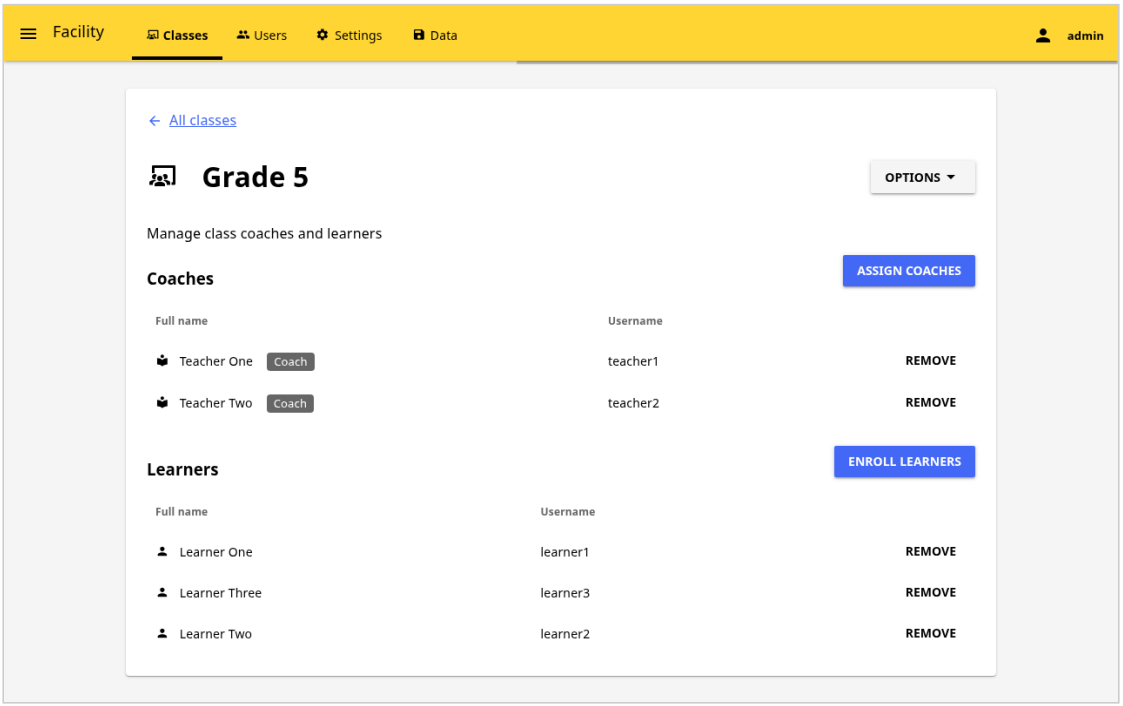


Figure 11. Learners in the class

3.4 Make a lesson and show it to the class

A good lesson mixes resources that explain (videos, pictures, activities) with resources that let learners practise (exercises). This section shows you how to make a lesson and make it visible to the class.

1. Open **Coach**, choose the class **Grade 5**, and click **Lessons > NEW LESSON**.

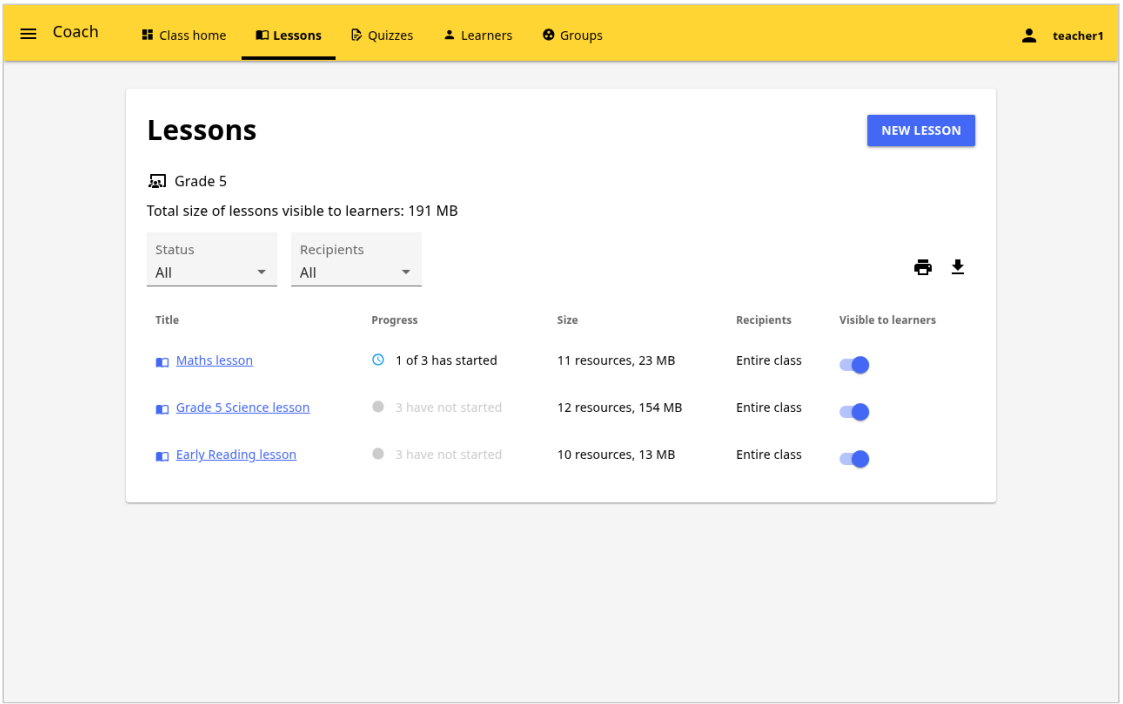


Figure 12. Lessons list

2. Type a title (`Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**. (In the demo we pressed **CANCEL**, and we use the ready-made **Maths lesson** in the steps below.)

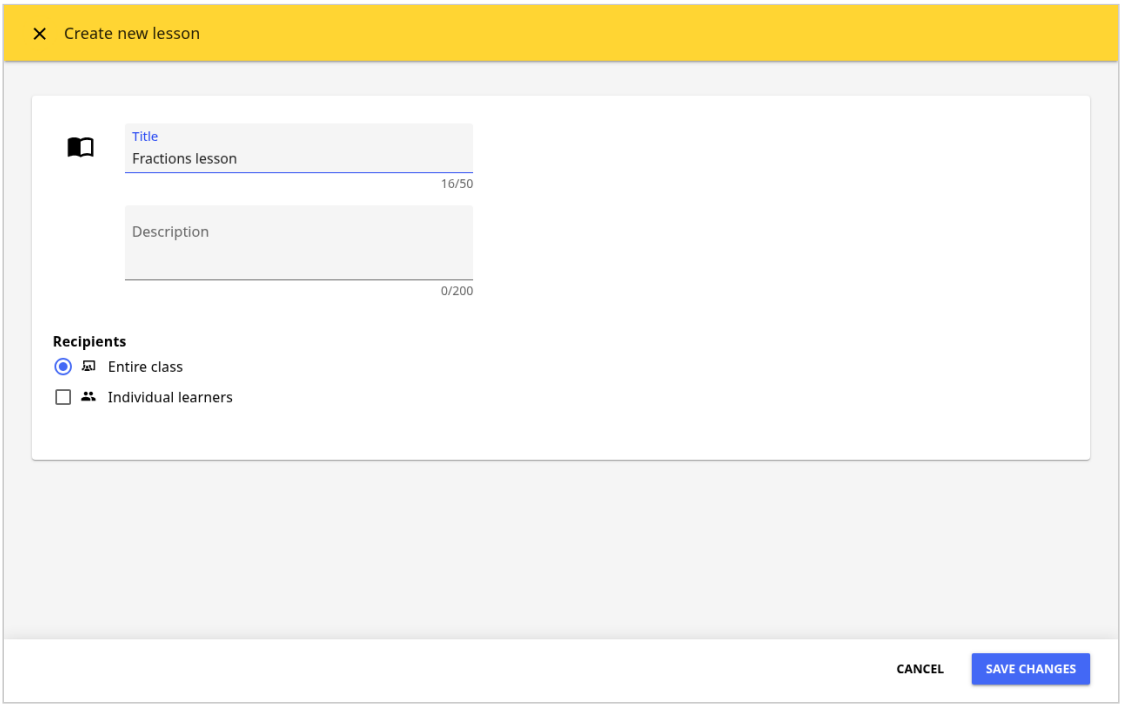


Figure 13. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

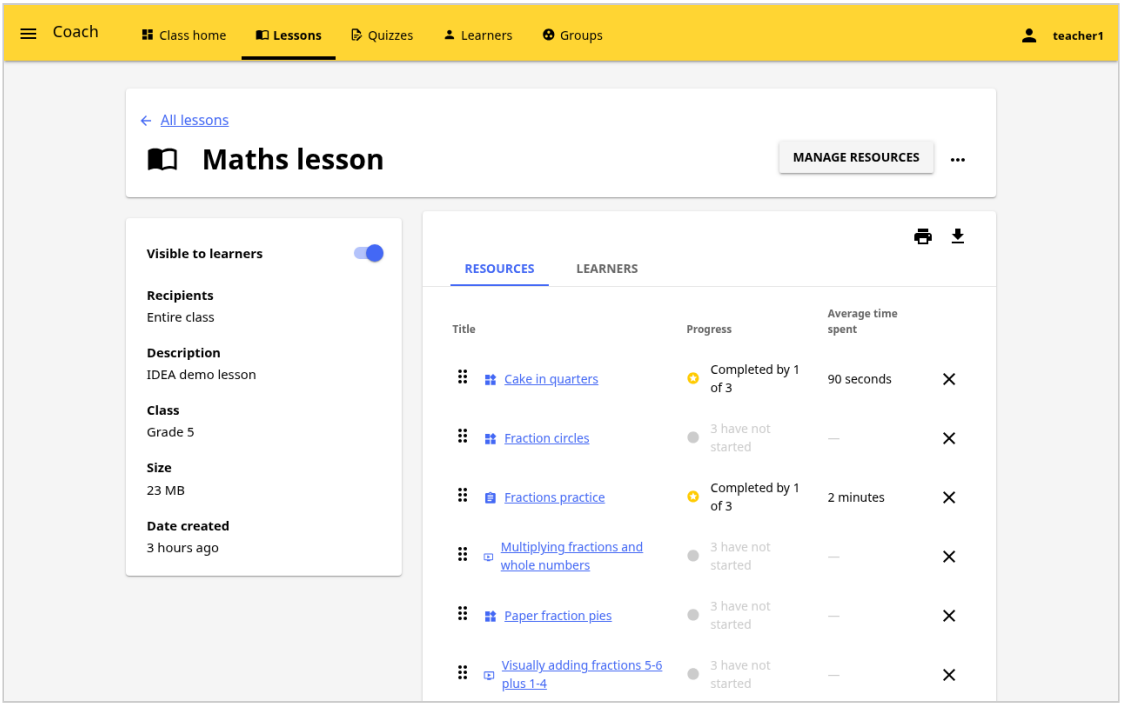


Figure 14. Maths lesson

4. The **Add resources to lesson** window opens. Choose the channel (**Maths**). You see its topics: **Fractions**, **Multiplication** and **Times Tables**. Open **Fractions**. You see seven items: three pictures (**Cake in quarters**, **Fraction circles**, **Paper fraction pies**), three short videos, and the exercise **Fractions practice**. Tick a mix, for example the three pictures, one video and **Fractions practice**. When you are done, click **SAVE & FINISH** (or the close **X**, if you did not mean to change anything). In the demo we opened the window and then closed it without adding ticks, because the **Maths lesson** already has these resources.

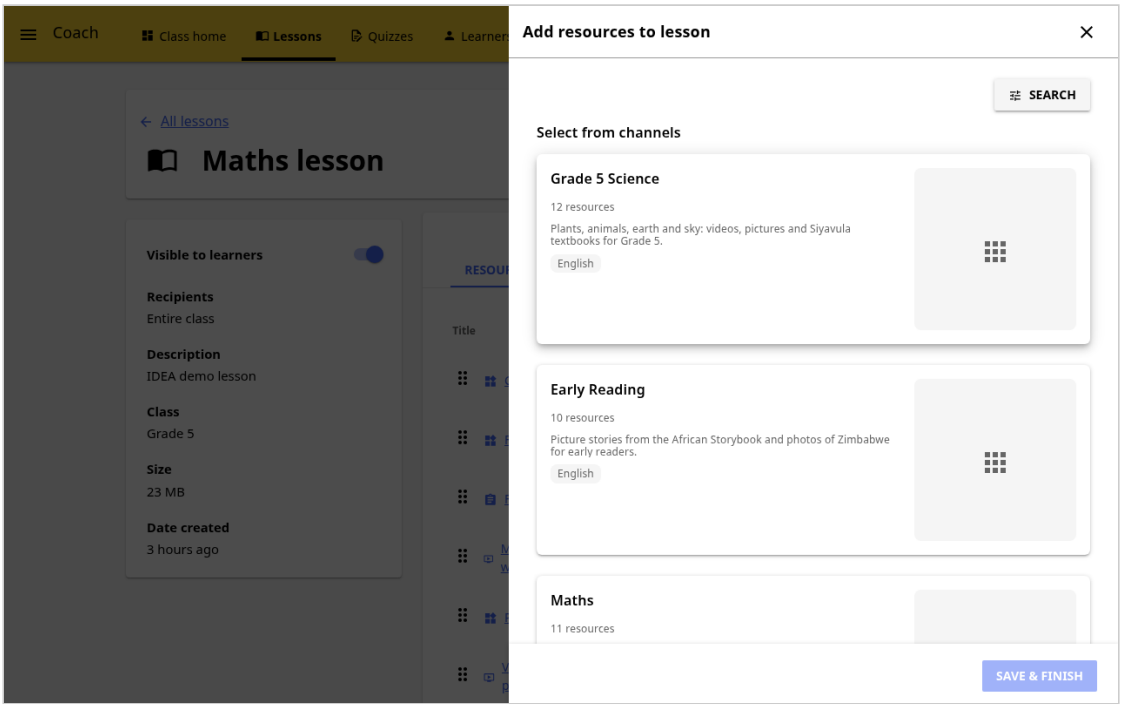


Figure 15. Add resources to lesson

5. Back on the lesson page, check the resources and switch on **Visible to learners**. The ready-made **Maths lesson** is already visible and already includes these Fractions items, together with the rest of the Maths channel (a multiplication video, two times-tables documents and **Times tables practice**).

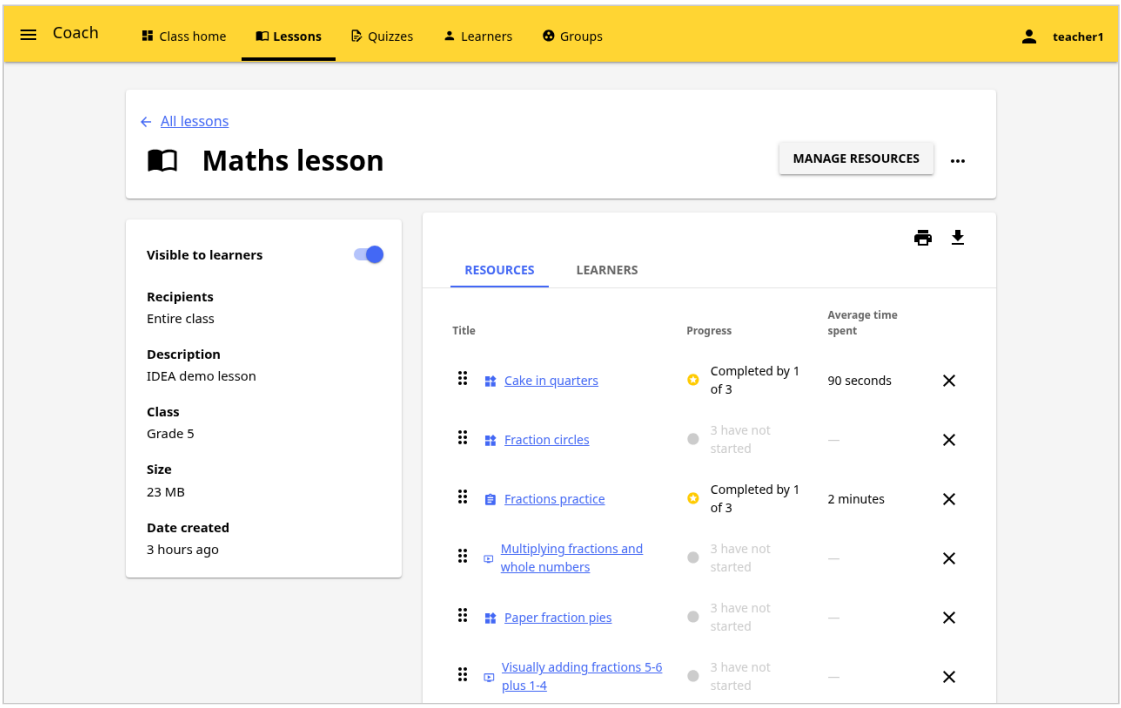


Figure 16. Lesson with its resources, visible

3.5 Make a quiz

Quizzes are built from exercises only. This section first lets you try a ready-made exercise and quiz in Kolibri Demo, then shows you how to make a quiz of your own.

Try it in Kolibri Demo

Kolibri Demo has exercises and a quiz ready for the class **Grade 5**: in **Maths**, **Fractions practice** and **Times tables practice** (the goal for each is **Get 4 correct**); in **Early Reading**, **Reading: Rudo and the goat**; and the quiz **Grade 5 Practice Quiz** (10 questions: 4 from Rudo and the goat, 6 from Fractions practice). Try them first as a learner, then see the results as the teacher. Use a second browser window (private window) for the learner.

A. A learner does an exercise

1. **Do:** sign in to Kolibri Demo as `learner1` (no password). Under **Recent lessons**, click **Maths lesson**.
   ✔ **Check: Home** shows the class **Grade 5**, three lessons and the quiz **Grade 5 Practice Quiz**.

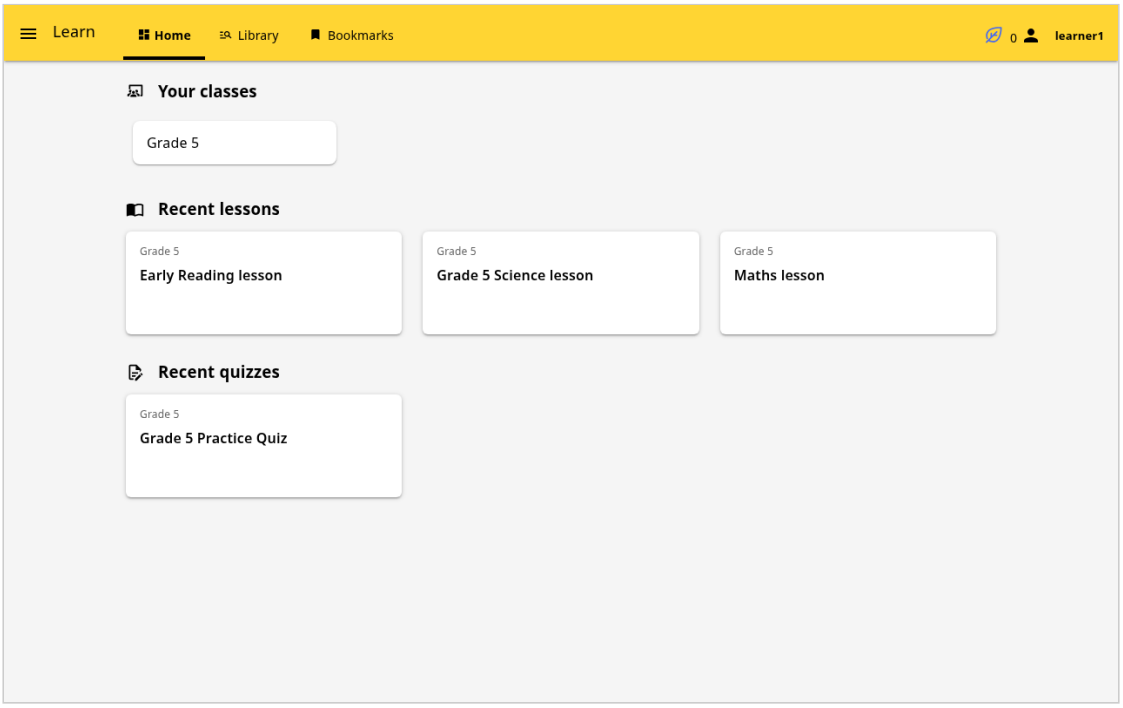


Figure 17. Learner home in Kolibri Demo

2. **Do:** in the lesson, click **Fractions practice** (marked **Practice**, with the goal **Get 4 correct**).
   ✔ **Check:** the lesson lists its items in order, starting with the Fractions pictures and **Fractions practice**, then videos, documents and **Times tables practice**.

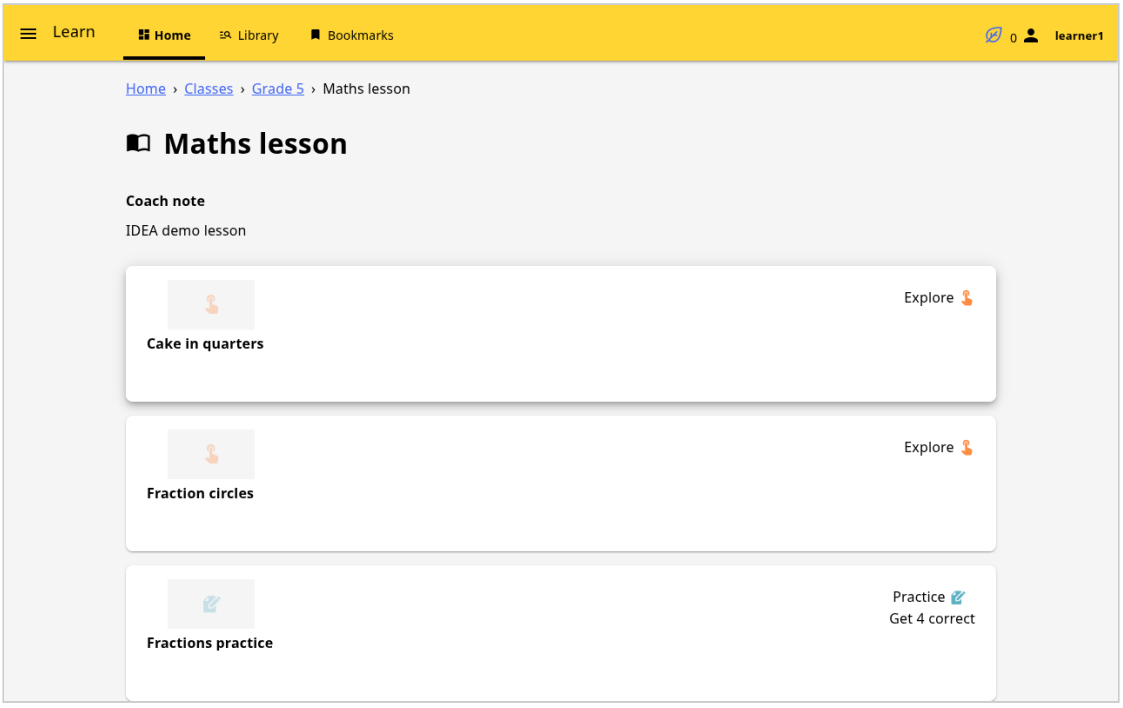


Figure 18. Maths lesson

3. **Do:** read the question, click an answer (or type a number), then click **CHECK**. After a right answer, go on to the next question; after a wrong answer, try again. Keep going until the goal **Get 4 correct** is reached.
   ✔ **Check:** the goal **Get 4 correct** is at the top, and **CHECK** is at the bottom. A green tick appears for each right answer, and the exercise is marked **Completed** when the goal is reached. In the Demo, `learner3` has already completed **Fractions practice**, so the screen below already shows the ticks and **Completed**.

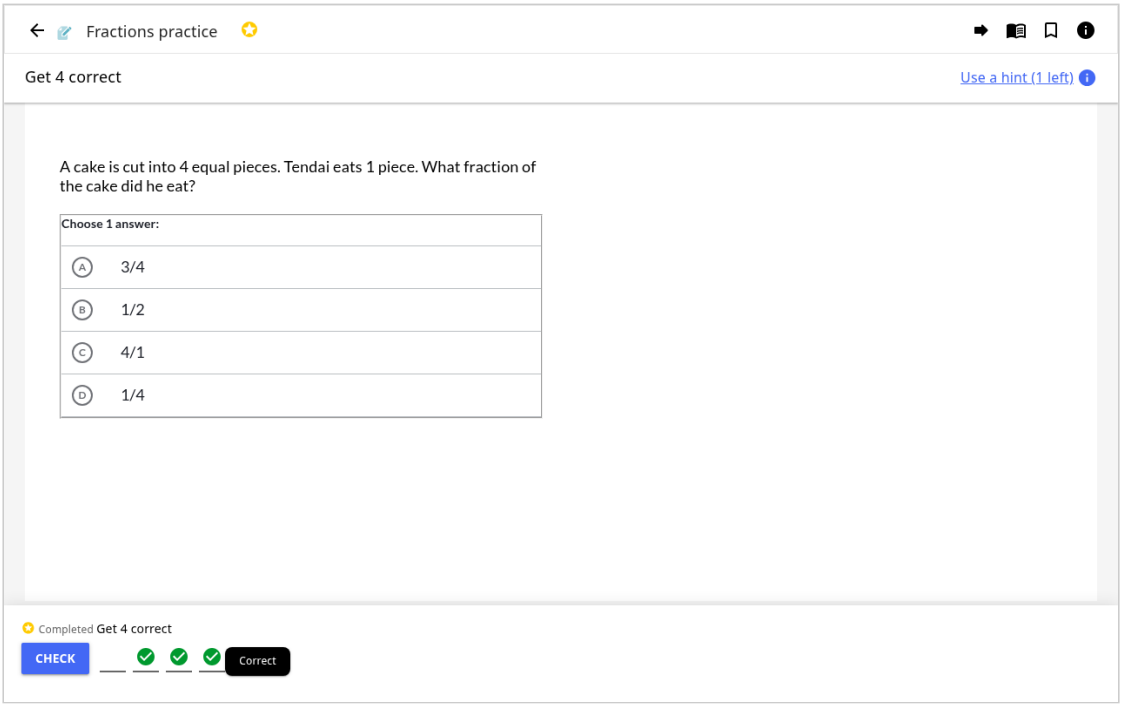


Figure 19. Exercise (Get 4 correct, CHECK)

B. A learner takes the quiz

1. **Do:** on **Home**, click **Grade 5 Practice Quiz** under **Recent quizzes**. Answer each question and move through them, then **SUBMIT QUIZ** and confirm. Answers cannot be changed after that.
   ✔ **Check:** in the Demo, `learner3` has already submitted the quiz. Opening it as `learner3` shows the result: **Completed**, score **70%** (7 out of 10), and an answer history you can look through question by question.

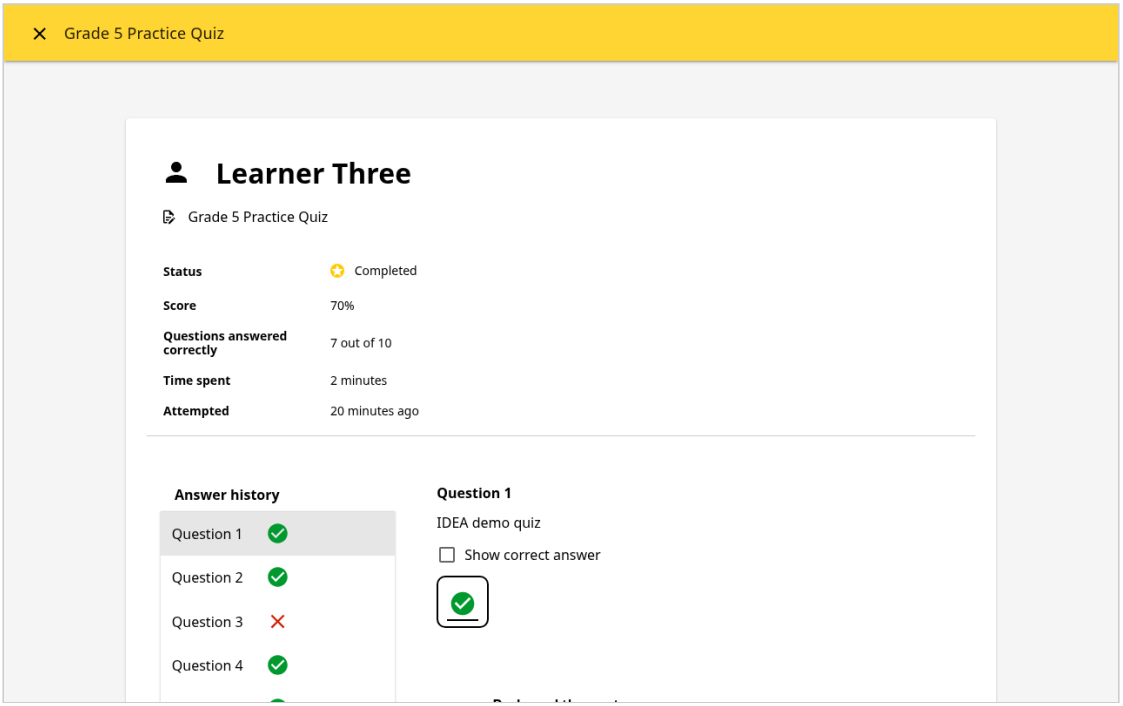


Figure 20. Quiz result

2. **Do:** go back to **Home**.
   ✔ **Check: Home** shows the quiz with its score (**Score: 70%** for `learner3`).

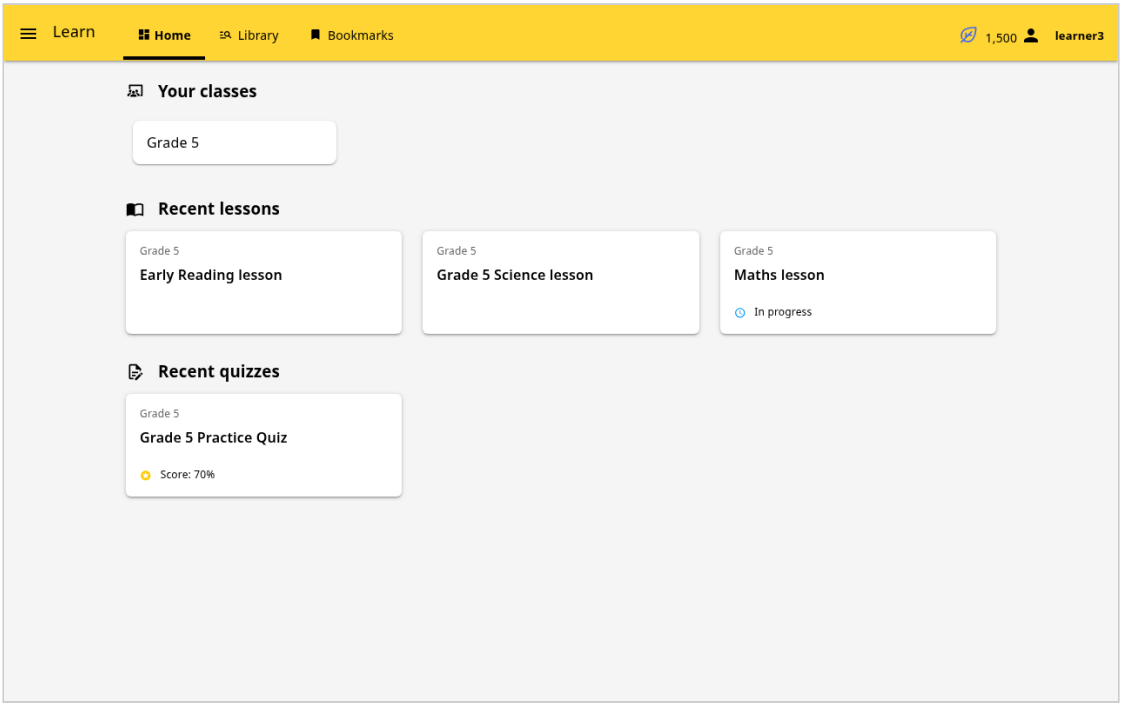


Figure 21. Quiz score on Home

C. The teacher sees progress

1. **Do:** in the teacher window, sign in to Kolibri Demo as `teacher1` and open **Coach > Grade 5** (the **Class home** tab).
   ✔ **Check: Class home** shows how many learners completed the quiz and started each lesson. **Class activity** lists what learners did.

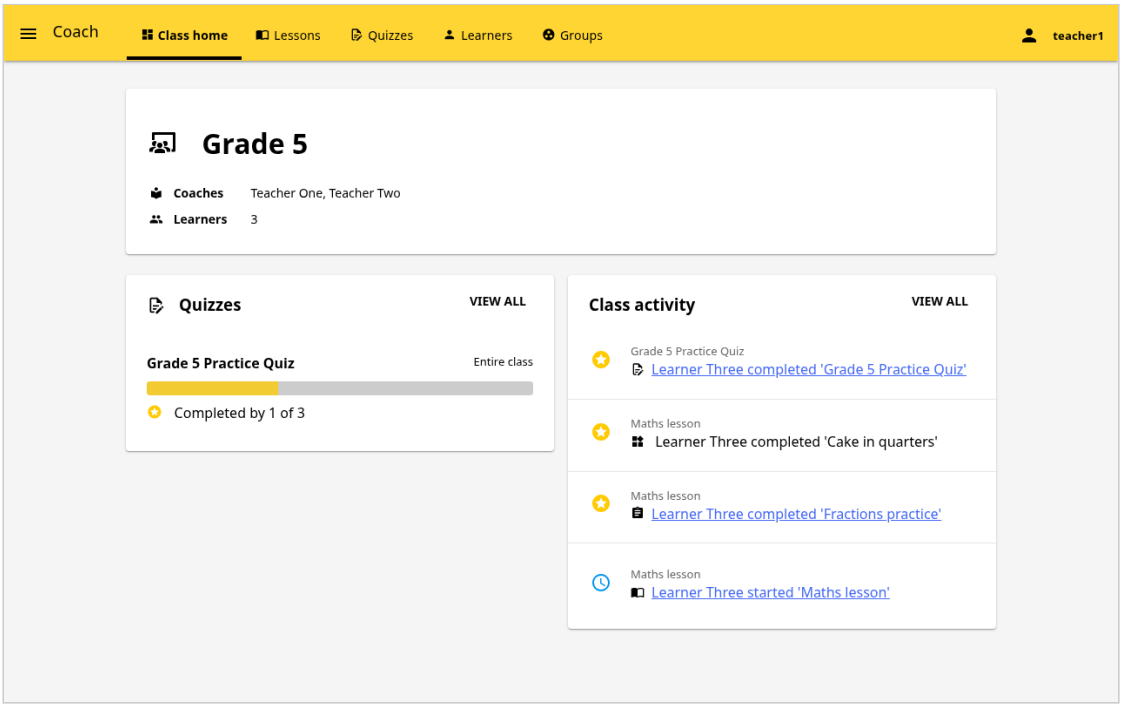


Figure 22. Coach class home

2. **Do:** click **Lessons > Maths lesson**, then the **Learners** tab on the lesson (the **Resources** tab shows time spent on each item).
   ✔ **Check:** each learner is listed as **Completed**, **Started** or **Has not started**.

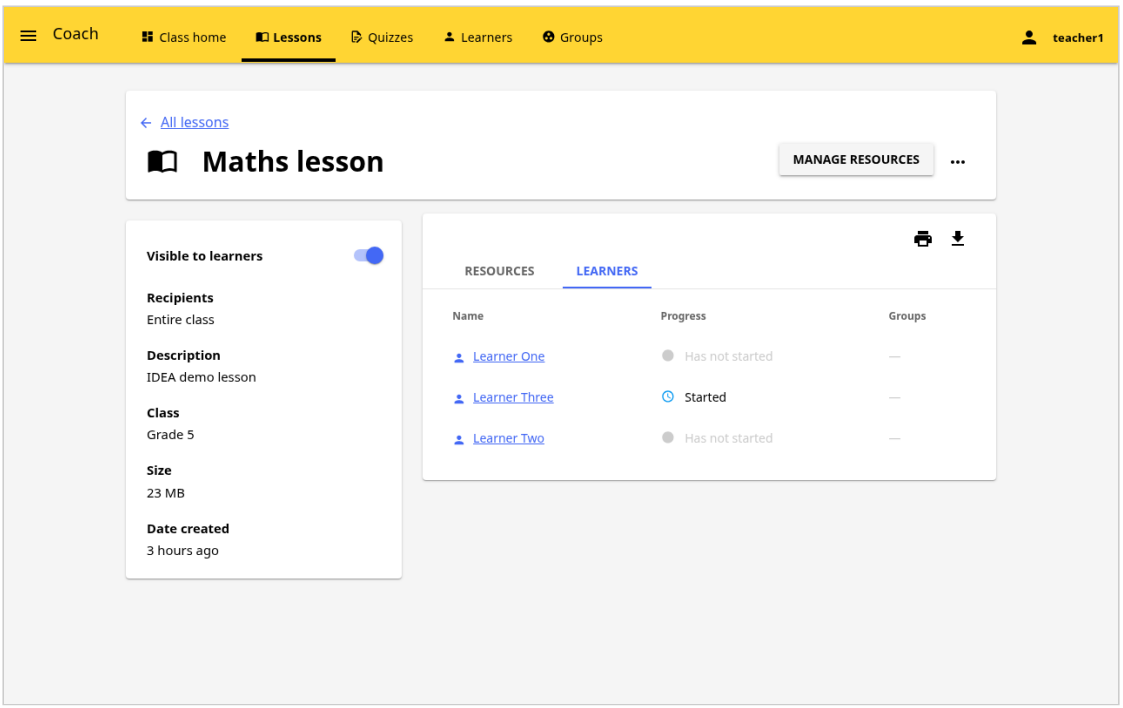


Figure 23. Lesson learners

3. **Do:** click **Quizzes > Grade 5 Practice Quiz**.
   ✔ **Check:** each learner's progress and score. The **Difficult questions** tab shows the questions many learners got wrong.

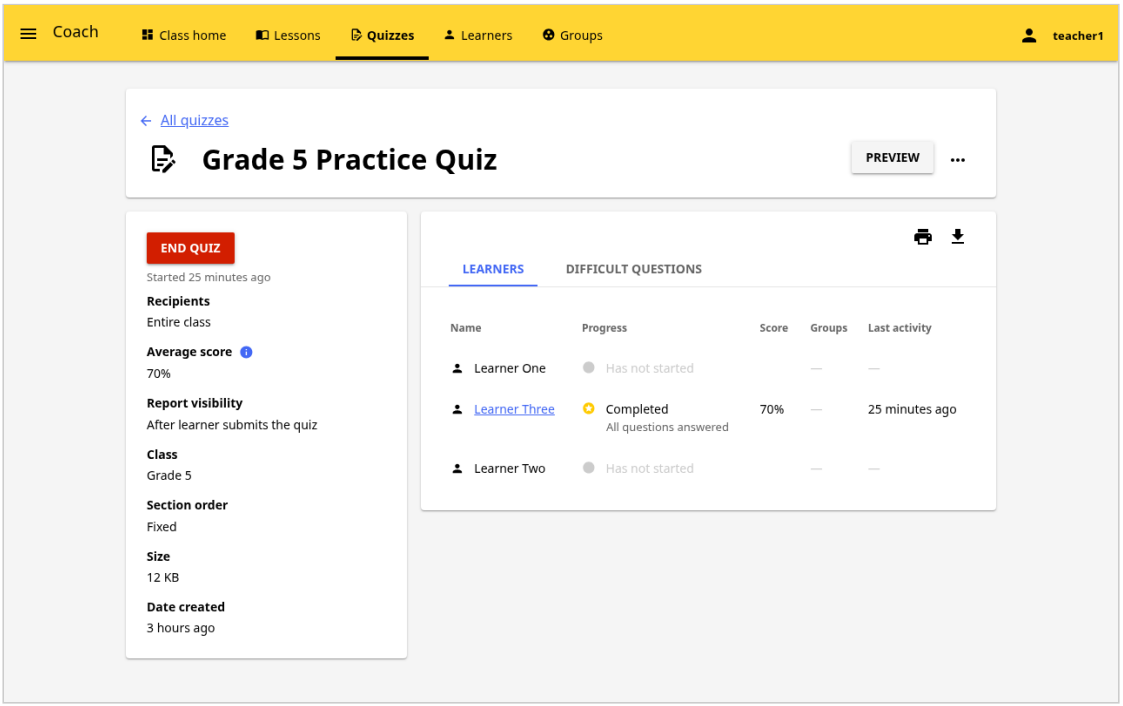


Figure 24. Quiz report

In Kolibri Demo, `learner3` has already done **Cake in quarters**, **Fractions practice** and the quiz, so the reports are not empty. 3.6 explains the reports in more detail.

Make your own quiz

The steps are the same in Kolibri Demo and in the school's own Kolibri. You pick exercises, and Kolibri can choose the questions, or you can choose them yourself.

1. Click **Quizzes > NEW QUIZ**, then choose **Create new quiz**.

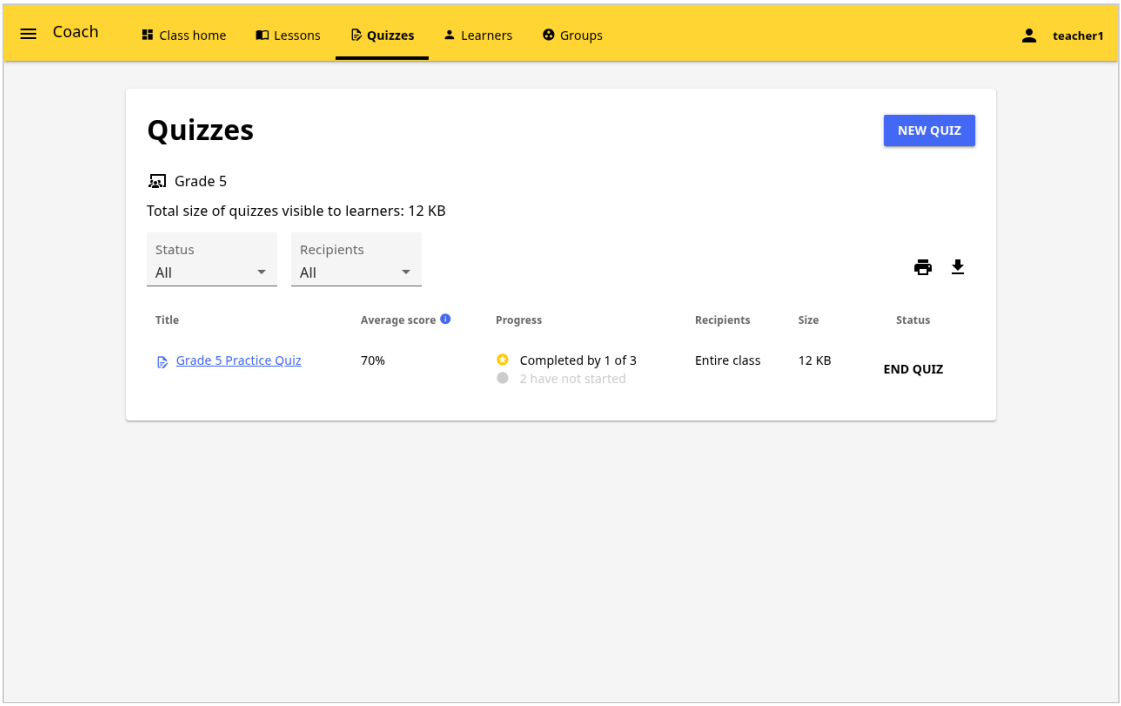


Figure 25. Quizzes tab

2. Type a title (`Fractions quiz`). Choose when learners may see their report, and keep **Recipients** as the entire class unless you want only some learners. A quiz can have one or more sections. Click **+ ADD QUESTIONS**, set how many questions you want, then **CONTINUE**. Browse to an exercise (for example **Maths > Fractions > Fractions practice**), tick it, and add the questions. You can review, replace or reorder questions, then **SAVE** or **SAVE & CLOSE**.
3. Check the questions and the section order (**Randomized** or **Fixed**).

4. A new quiz is not open to learners yet. When the class is ready, click **START QUIZ**. When the class has finished, open the quiz again and click **END QUIZ**, so no more answers come in. In the Demo, **Grade 5 Practice Quiz** is already started, so its status on the Quizzes list shows **END QUIZ**.

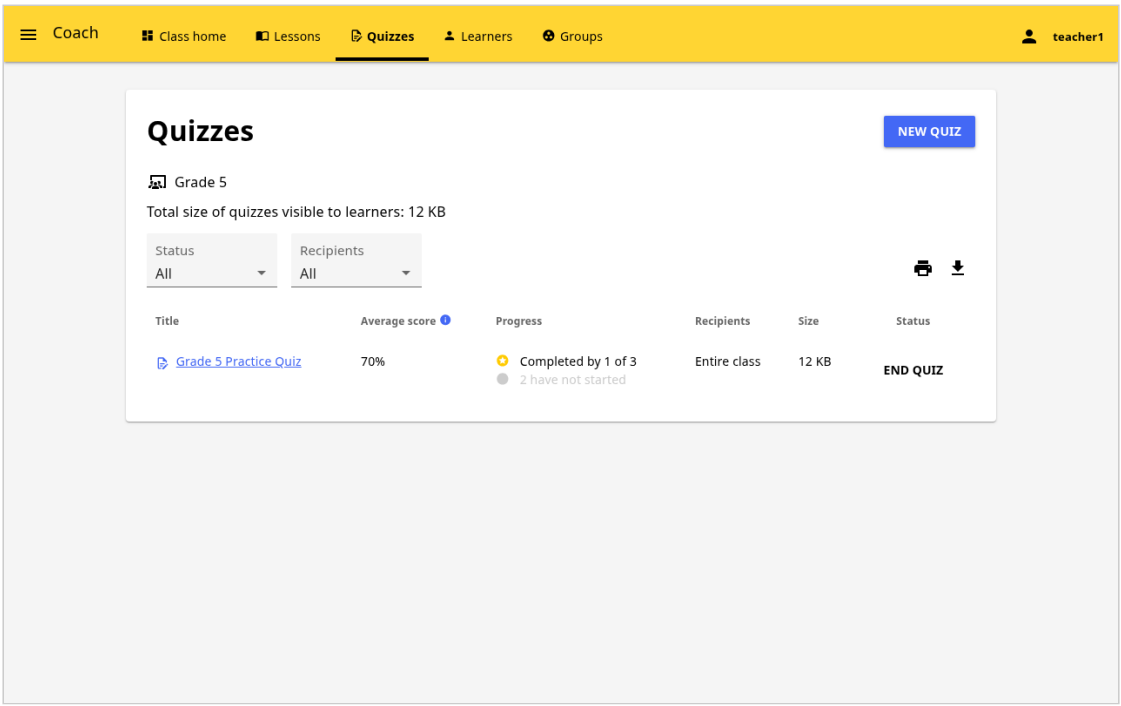


Figure 26. Quiz started, status END QUIZ

3.6 See progress (reports)

This is how you read what the class has done, after learners have worked (chapter 4).

In Kolibri 0.19 you read reports on the lesson, the quiz or the learner, rather than on a separate **Reports** tab.

1. Open **Coach > Grade 5**. Use **Lessons**, **Quizzes** and **Learners** (and **Class home** for the overview).

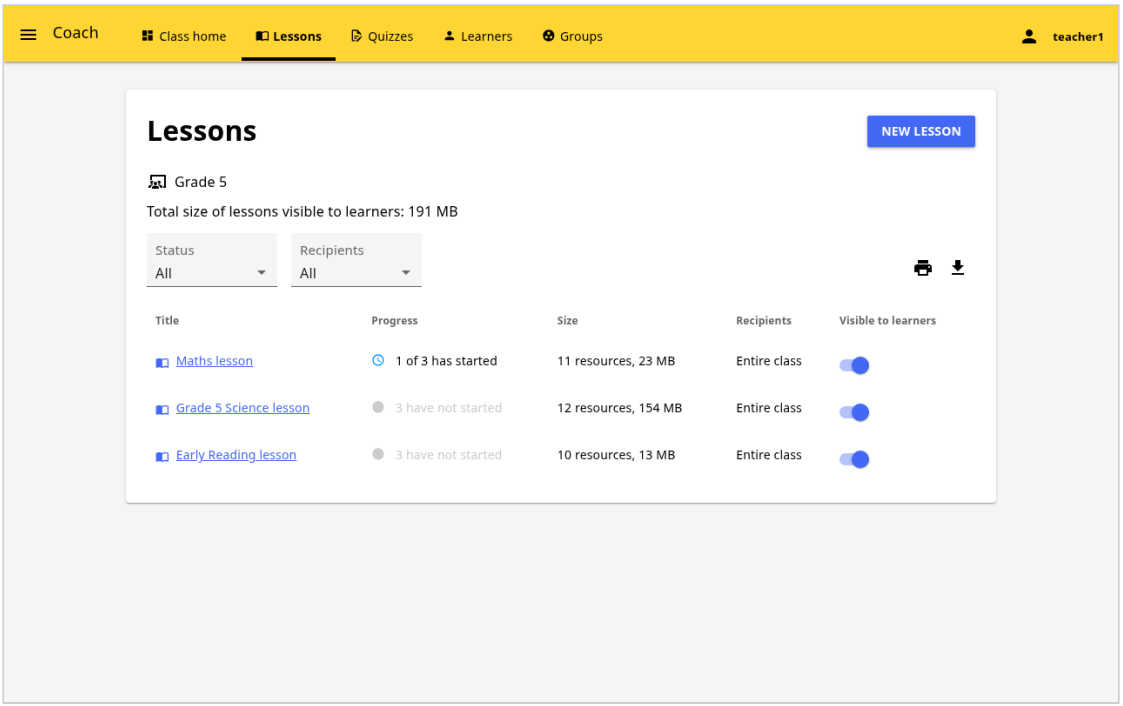


Figure 27. Lessons with progress

2. **Lessons > (the lesson) > Learners**: who completed, started or did not start.

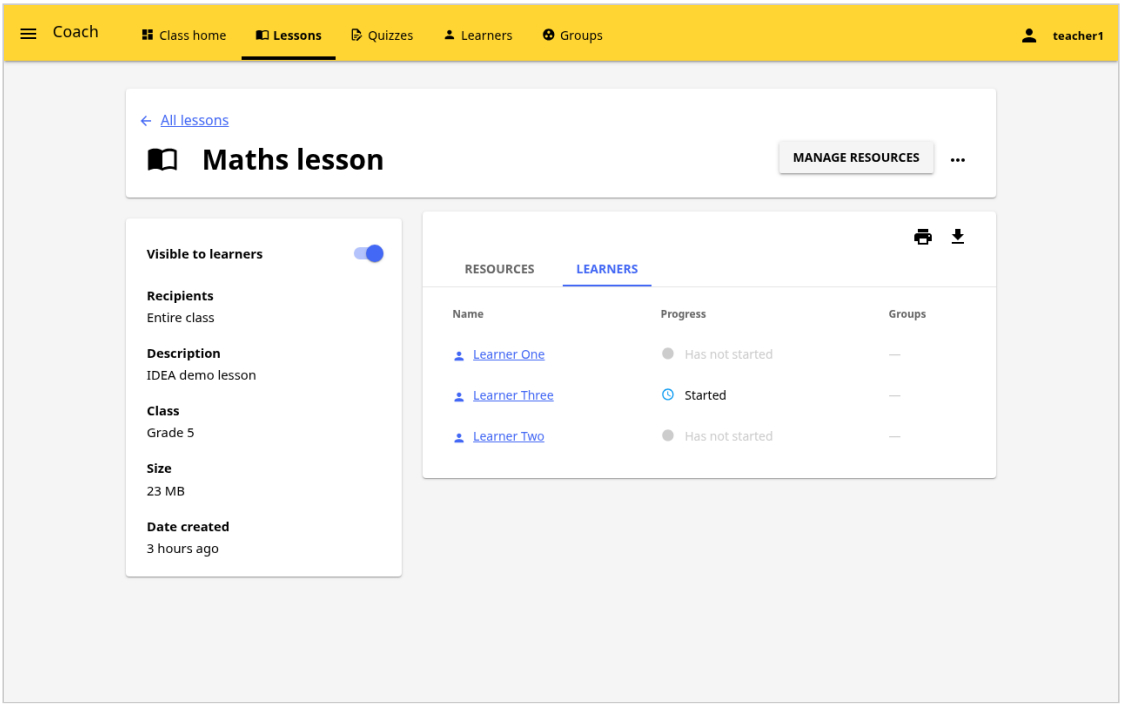


Figure 28. Lesson report

3. **Quizzes > (the quiz)**: status and score per learner; **Difficult questions** shows what to re-teach.

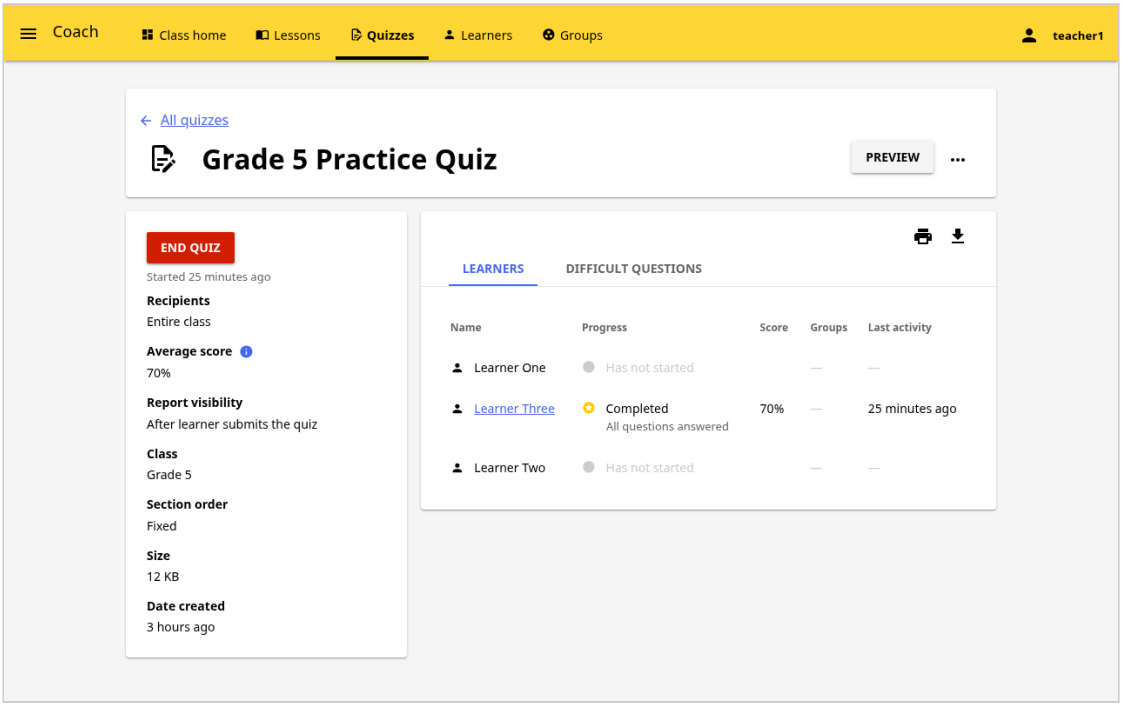


Figure 29. Quiz report

4. **Learners > (a name)**: that learner's lessons and quizzes. A lesson that is hidden from learners can still appear on this report.

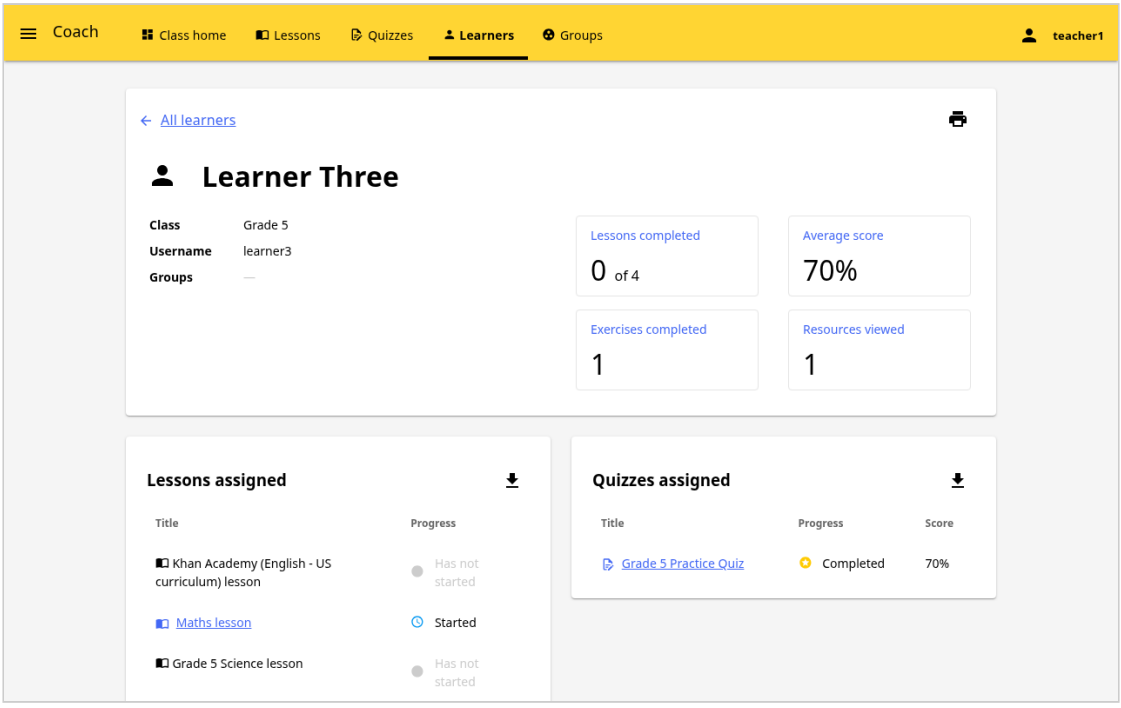


Figure 30. Learner report

Reports can be printed or exported as CSV (icons top right).

> **Tip.** Do not chase 100%. Look for the question most learners got wrong and teach it again.

What the lesson report means

Kolibri shows	What it means	What to do
Completed	The learner finished the resource or the lesson	Nothing. Give them the next lesson or a quiz
Started	The learner began but did not finish	They can finish next time. Note the name
Has not started	The learner did not open it	Ask them. Often they did not sign in with their own name, or the lesson was not visible to learners

You do not need every learner at 100% after one class. Check that they signed in, worked through the explaining resources and tried the exercises.

3.7 Teaching one class period

This puts 3.4, 3.6 and chapter 4 together for one lesson. In Kolibri Demo, use `teacher1`, the class **Grade 5** and the **Maths lesson**.

Before class

1. **Do:** sign in as `teacher1`, open **Coach > Grade 5 > Lessons** and click **Maths lesson** (3.4).
   ✔ **Check:** the resources are in the order you want: explaining resources (pictures, videos, documents) first, exercises where you want learners to practise.
2. **Do:** check **Recipients**.
   ✔ **Check:** it shows **Entire class**.
3. **Do:** switch on **Visible to learners** (it is already on for the Maths lesson).
   ✔ **Check:** the switch is on. Learners see the lesson the next time they open **Home**.

During class

1. **Do:** ask every learner to open Kolibri Demo and sign in with **their own** username (chapter 4).
   ✔ **Check:** each learner sees **Maths lesson** on **Home**. A learner who uses someone else's name, or does not sign in, does not appear in the reports.
2. **Do:** tell the class to go through the explaining resources first, then do the exercises, in the order of the lesson.
3. **Do:** walk the room. Help learners who are stuck, but do not do the exercises for them.

After class

1. **Do:** open **Lessons > Maths lesson** (3.6).
   ✔ **Check:** each learner is listed as **Completed**, **Started** or **Has not started**. The table in 3.6 says what to do about each.
2. **Do:** pick the one thing most learners found hard and teach it again next time. Do not chase 100%.

If something is wrong

What you see	What to do
The lesson is not on a learner's screen	Check that **Recipients** shows the class and that **Visible to learners** is on
A learner cannot sign in	Check that they type their own username. If it still does not work, tell your school operator
Reports is empty	Learners must sign in with their own name before they start
A resource is missing from the lesson	Tell your school operator
The Kolibri card in the Console has no **Open**	Tell your school operator. Do not try to fix it yourself

4. Kolibri for learners (try it yourself)

This chapter is for you to see Kolibri as your class sees it, so you can show them the same steps.

Your learners open Kolibri the same way you do: on a classroom computer they double-click **IDEA** on the desktop and click **Open** on the Kolibri card in the Console (2.2.1). Show them this once. To try it yourself with the demo learners, open **Kolibri Demo**.

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. Learners have no password in Kolibri. They sign in with only their username, for example `learner1`.

> **On every IDEA box**, both **Kolibri** and **Kolibri Demo** already let learners sign in without a password. **At an existing school**, a teacher with admin rights (or the Kolibri admin) turns this on in Kolibri: open **Facility > Settings**, under **How learners sign in** choose **Enter username only**, then click **SAVE CHANGES**.

If a welcome window appears the first time you open Kolibri Demo, close it to go on.

4.1 Do a lesson

This is how a learner opens a lesson and works through it.

1. Sign in as `learner1`. **Home** shows **Your classes**, **Recent lessons** and **Recent quizzes**.

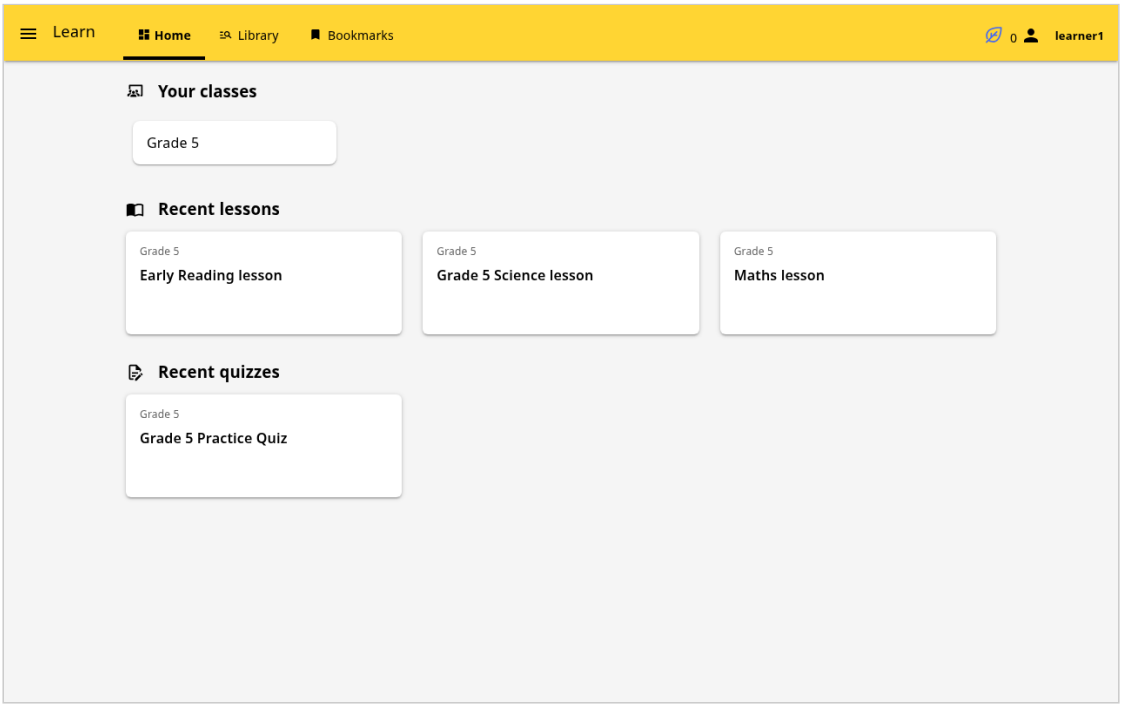


Figure 31. Learner home

2. Click **Maths lesson**. Its items are listed in order, starting with the Fractions pictures and **Fractions practice**.

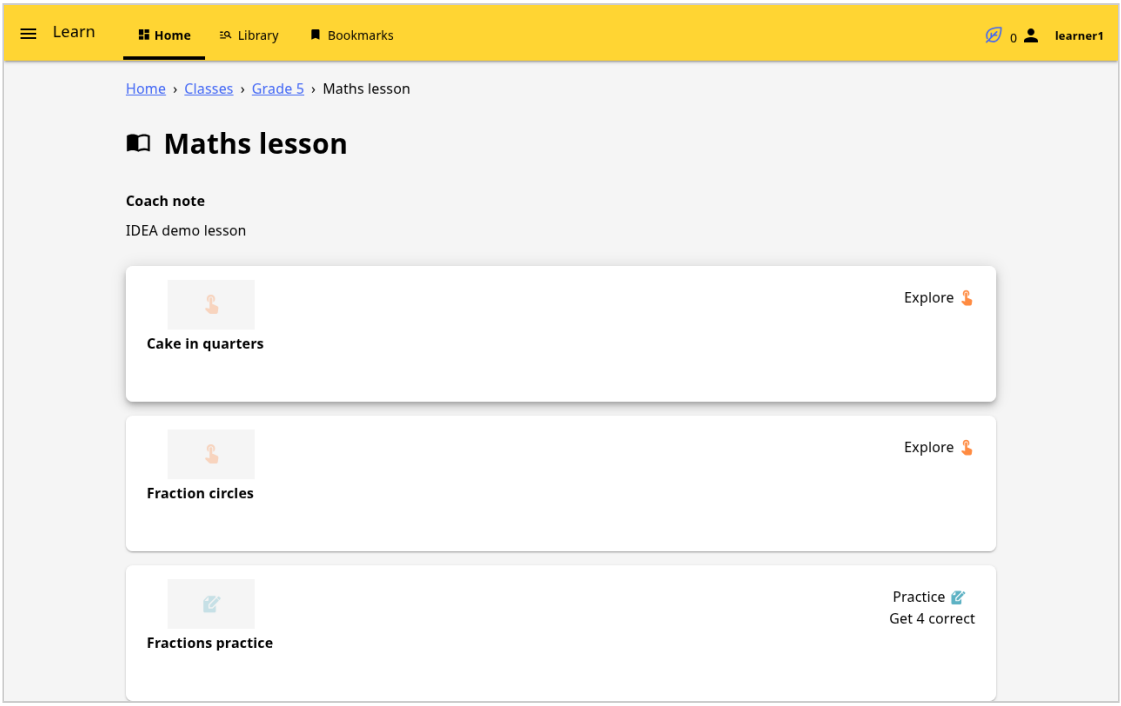


Figure 32. Lesson

3. Open an item, for example the first one (**Cake in quarters**, a picture activity). Videos open the same way.
4. Exercises work like this (in Kolibri Demo, try **Fractions practice** in the **Maths lesson**, or look at it as `learner3`, who has already completed it; see 3.5): type an answer or choose one, click **CHECK**. Keep going until the goal at the top (**Get 4 correct**) is reached.

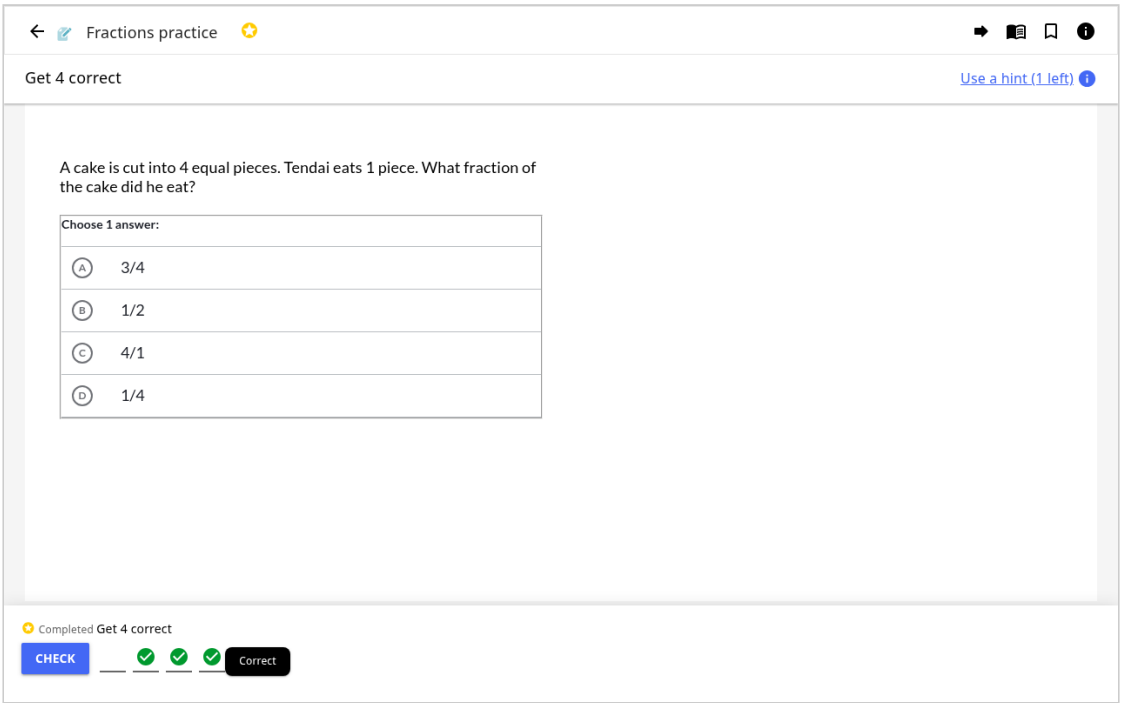


Figure 33. Exercise

5. Sign out. In the Demo, `learner3` has already completed **Cake in quarters** and **Fractions practice**, so the teacher reports already show a contrast with `learner1` and `learner2`.

4.2 Take a quiz

This is how a learner takes a quiz that has been started. In Kolibri Demo, use **Grade 5 Practice Quiz** (see 3.5).

1. Open the quiz from **Recent quizzes** (or the class page). Answer each question, then click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed. In the Demo, sign in as `learner3` to see a quiz that has already been submitted: the score is **70%** (7 out of 10) and you can look back at each question.

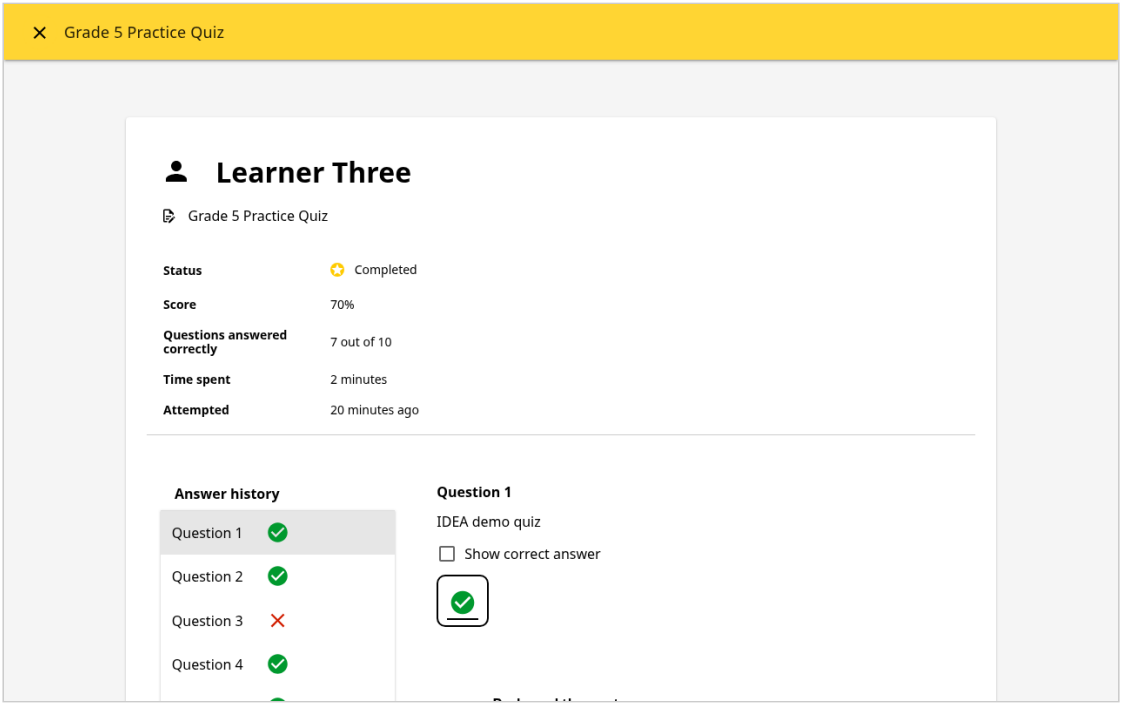


Figure 34. Quiz result

2. Back on **Home**, the quiz shows its score.

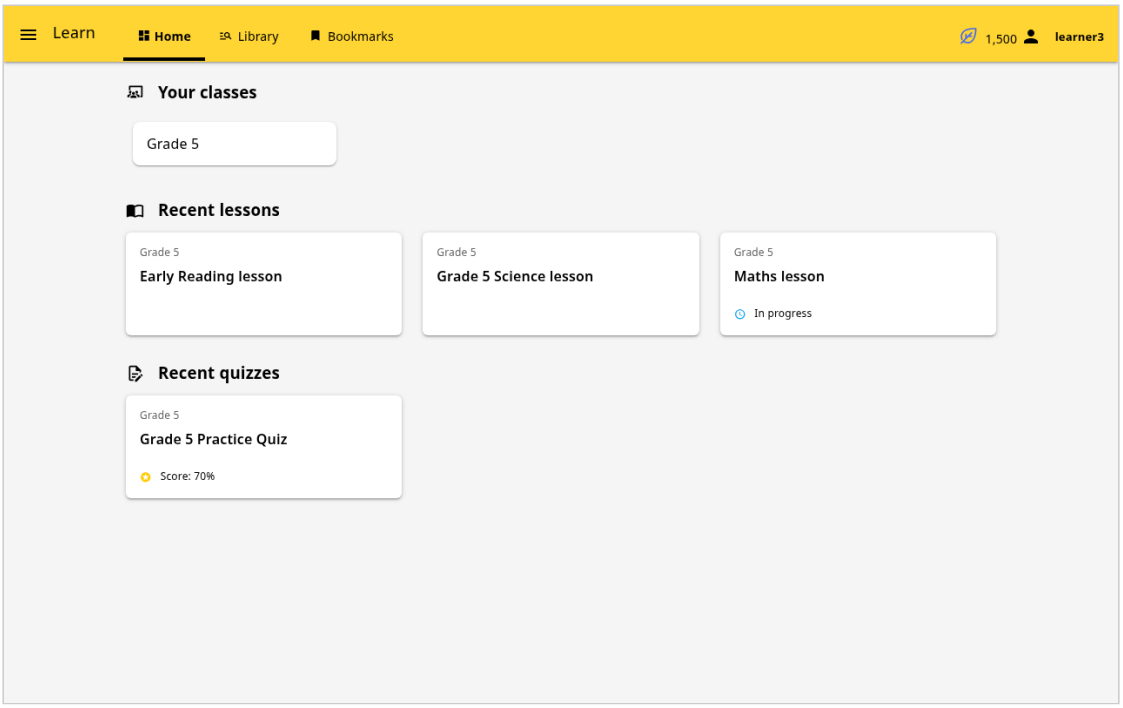


Figure 35. After submit, on Home

Now go back to the teacher window and look at the reports (3.6).

5. Nextcloud for teachers

About the examples in this chapter. Every IDEA box comes with the demo setup switched on: the app **Nextcloud Demo**. It has the demo teachers `teacher1` and `teacher2` (group `teachers`), the demo learners `learner1` to `learner3` (group `learners`), and in each teacher's files the folders `Worksheets` and `Teacher Notes`. The screens in this chapter call the class group `Grade 5`: in Nextcloud Demo, pick `learners` wherever the steps say the class group. The steps work the same at your school, but your names will be different. The demo accounts are listed in Appendix: demo accounts.

Office documents open in **Nextcloud Office**.

Open **Nextcloud Demo** from the Console (chapter 2) and sign in as `teacher1`. Click **Files** at the top.

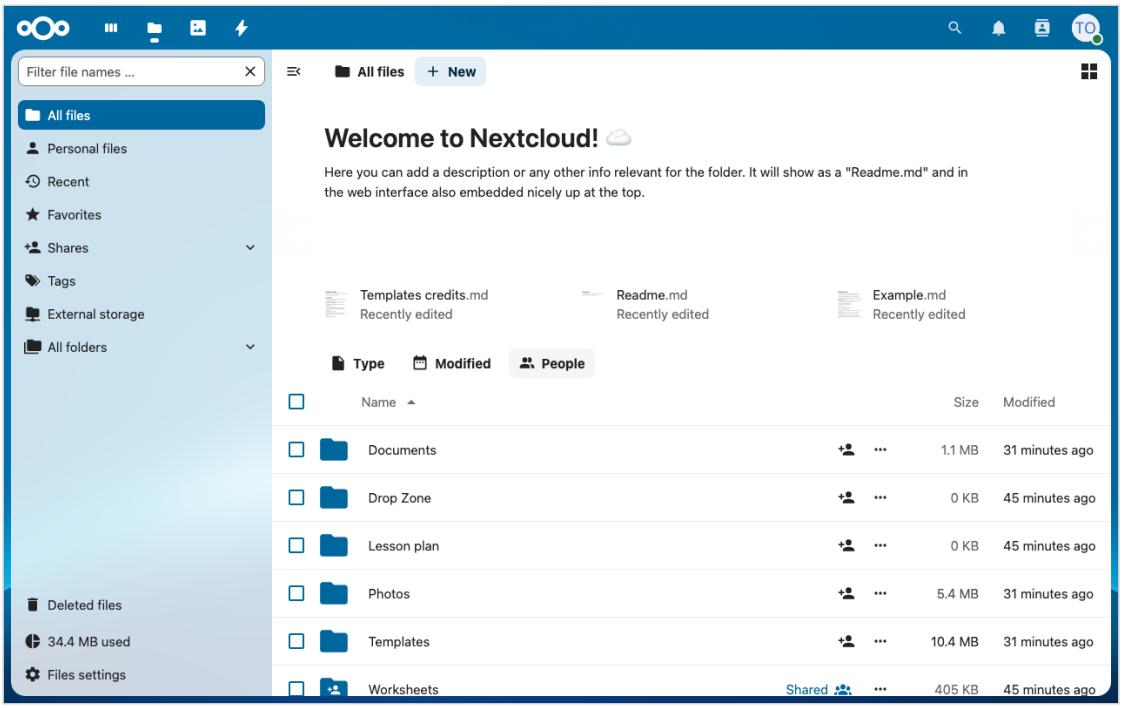


Figure 36. Files

5.1 Share worksheets with the class (view only)

1. Next to the folder `Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

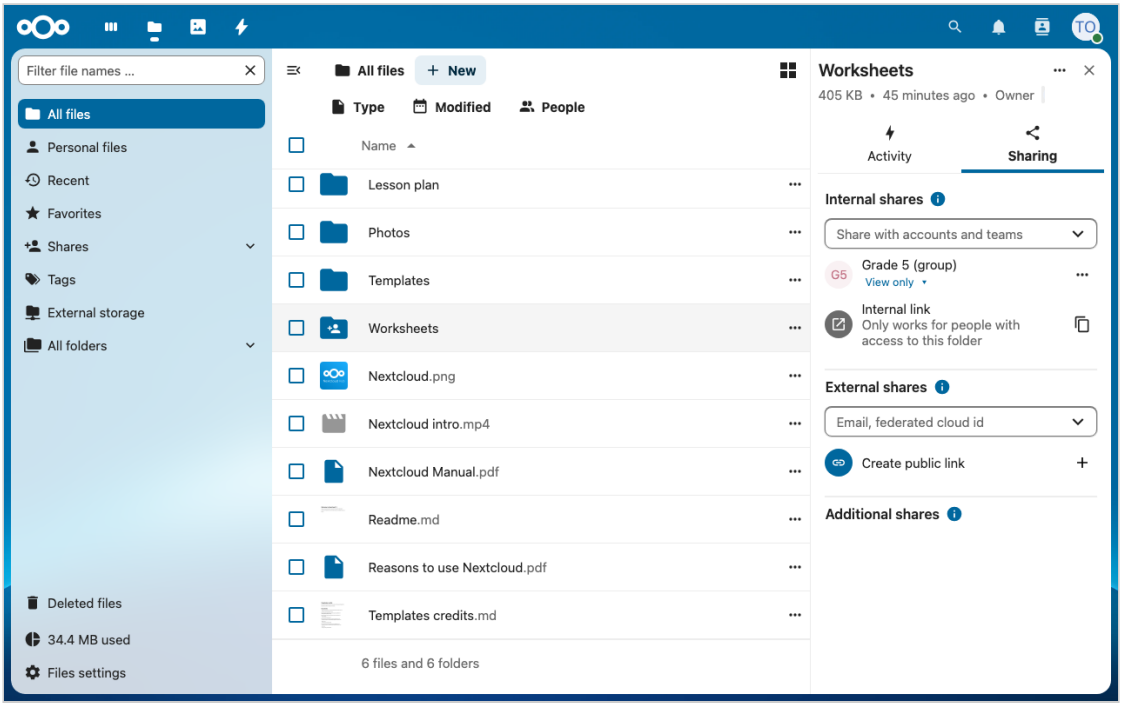


Figure 37. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type the class group (on the kit: `learners`) and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

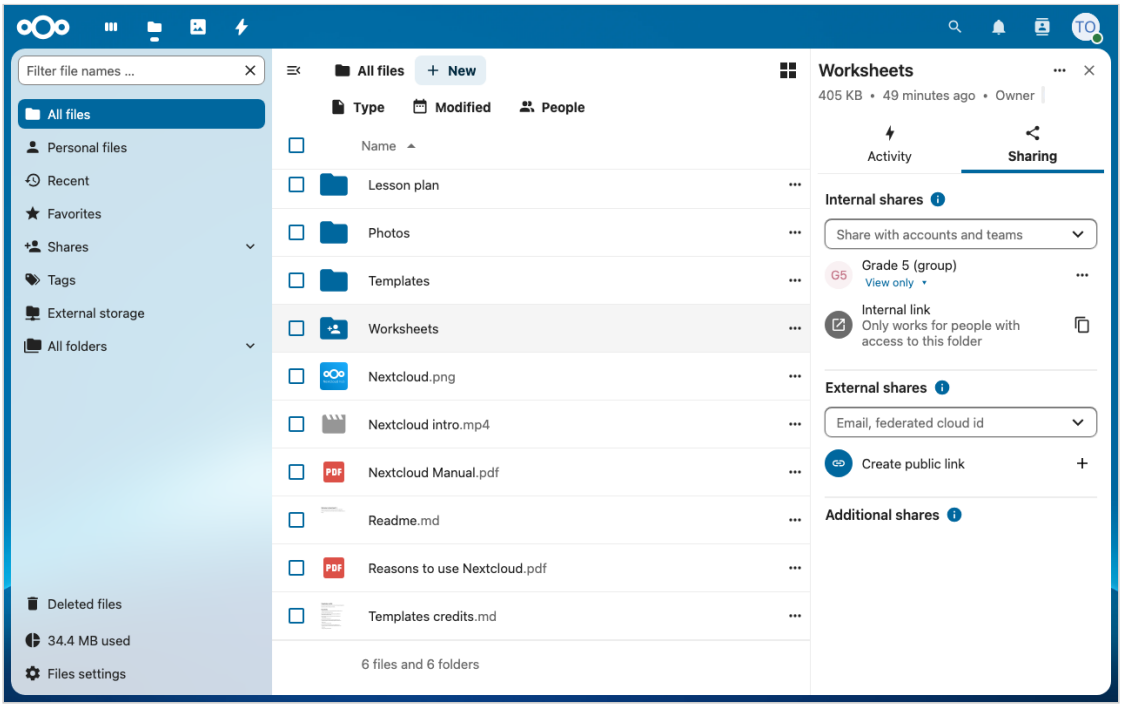


Figure 38. Shared with group

Learners now see the folder but cannot change or delete files.

5.2 Write a document together

1. Click **+ New > New document**.

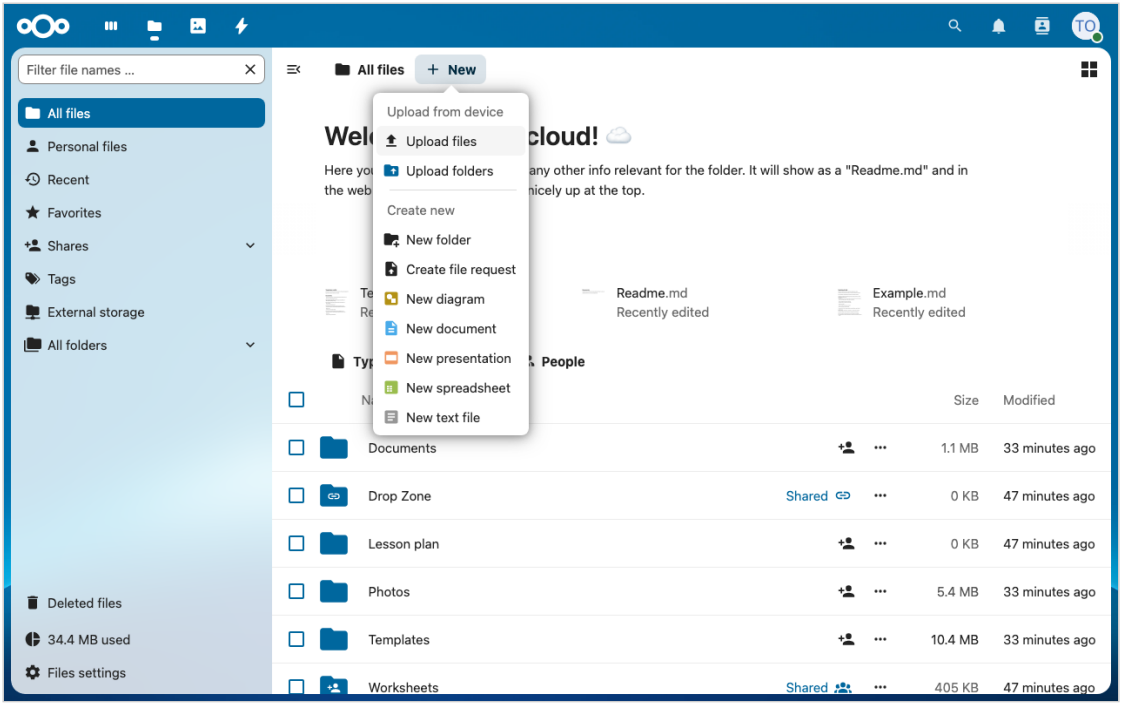


Figure 39. New menu

2. Type a name (`Lesson plan`) and click **Create**.

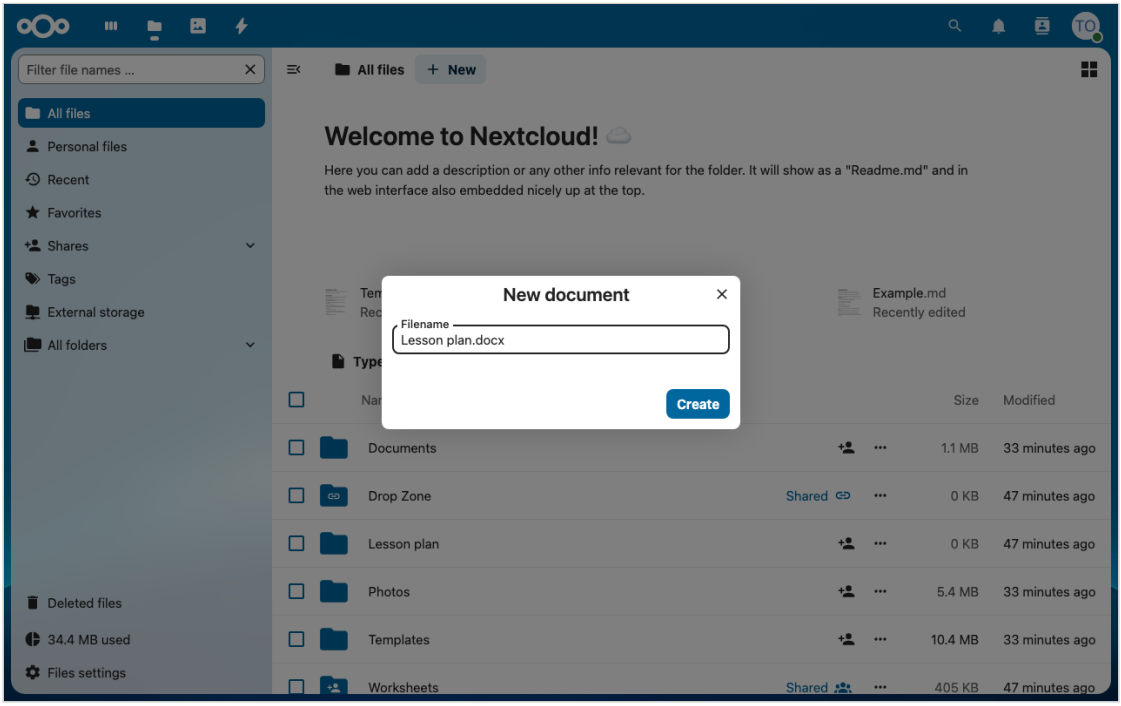


Figure 40. Name

3. Pick a template (Blank is fine) and click **Create**.

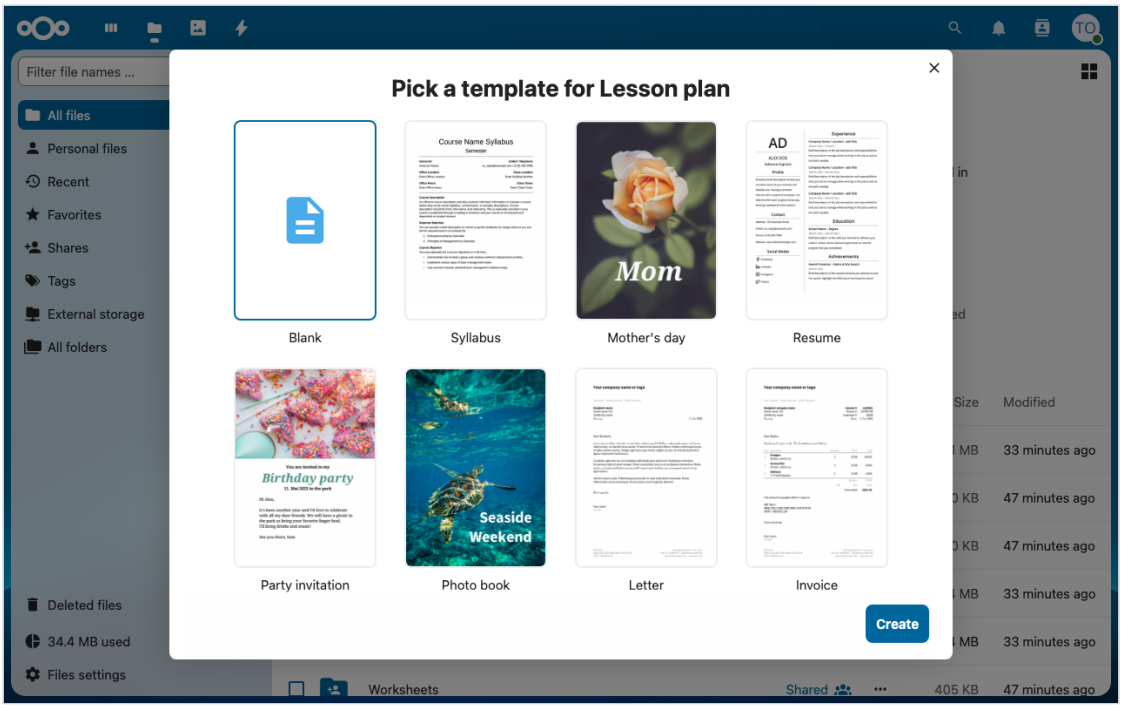


Figure 41. Template

4. Share the file with `teachers` and/or your class group (on the kit: `learners`), with editing allowed.

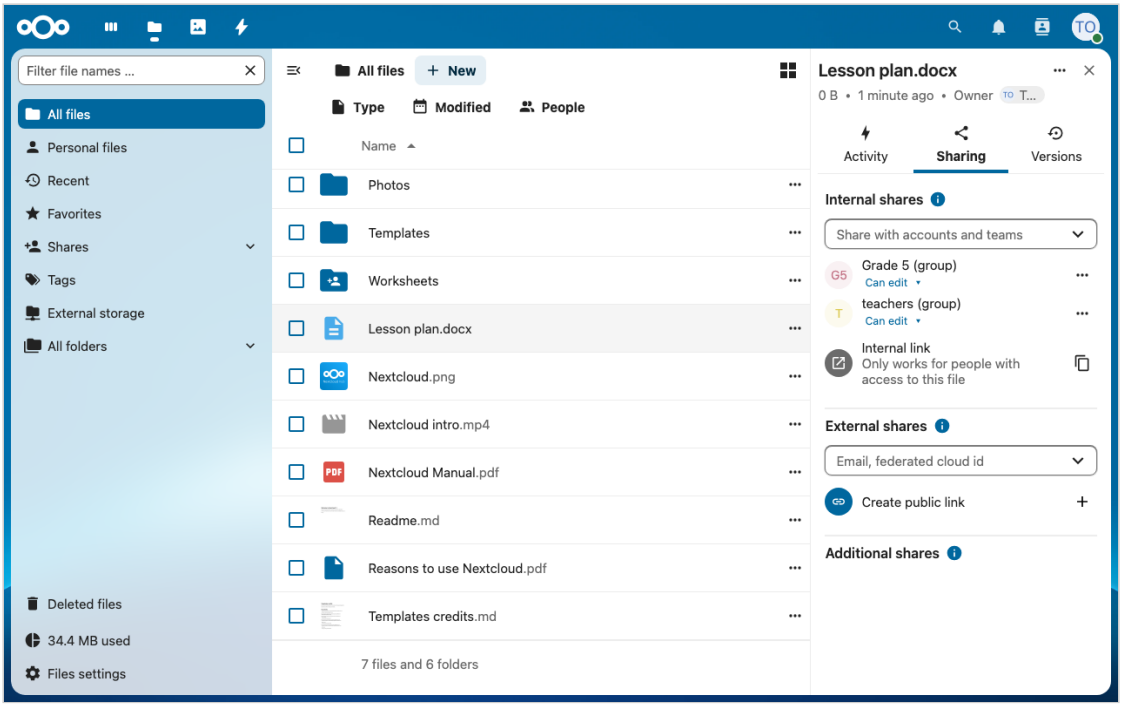


Figure 42. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

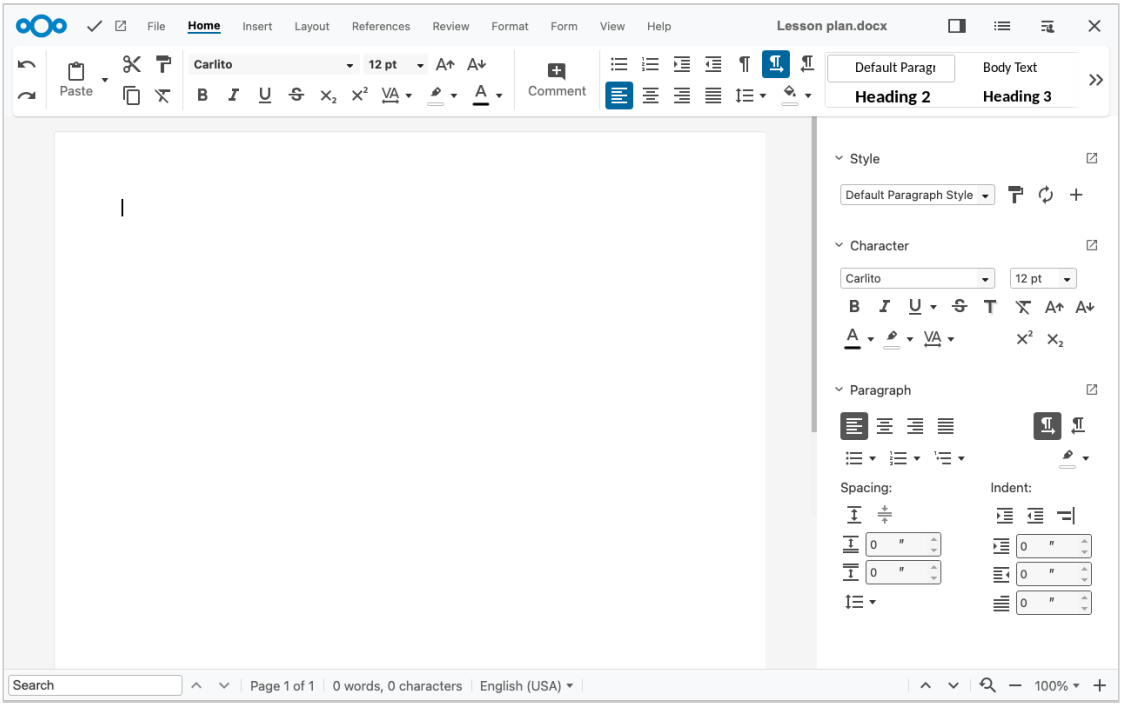


Figure 43. Nextcloud Office

5.3 Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Make a folder for the homework: **+ New > New folder**, name it `Drop Zone`. Then open sharing for the folder `Drop Zone`.

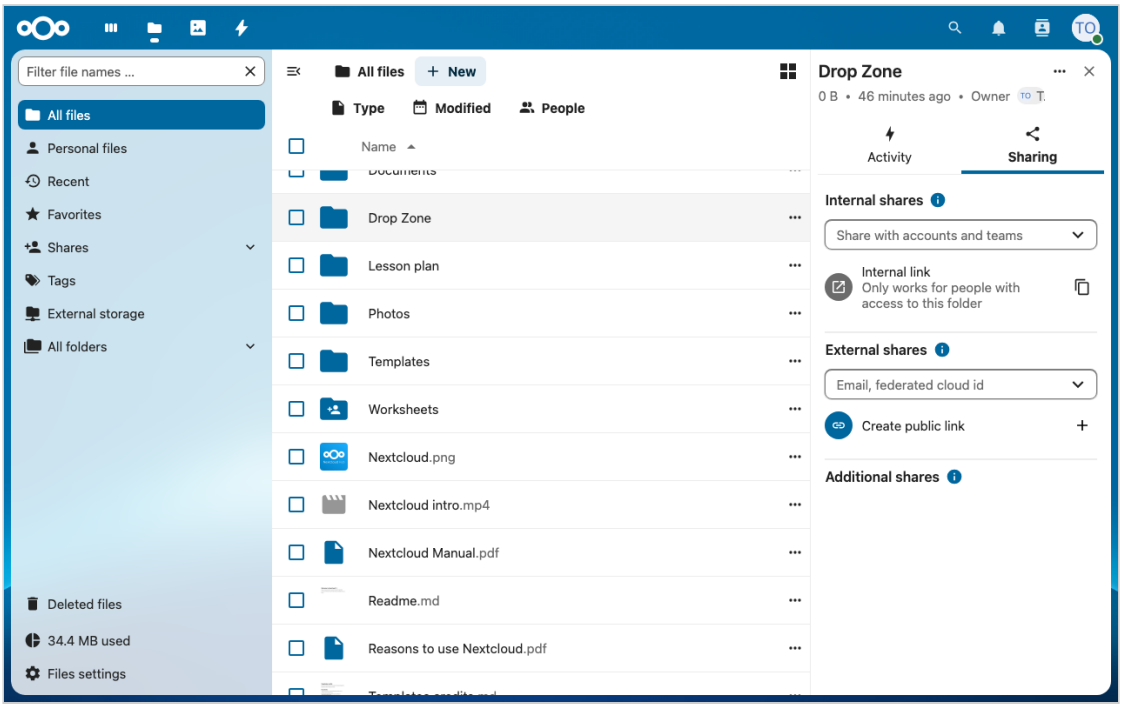


Figure 44. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

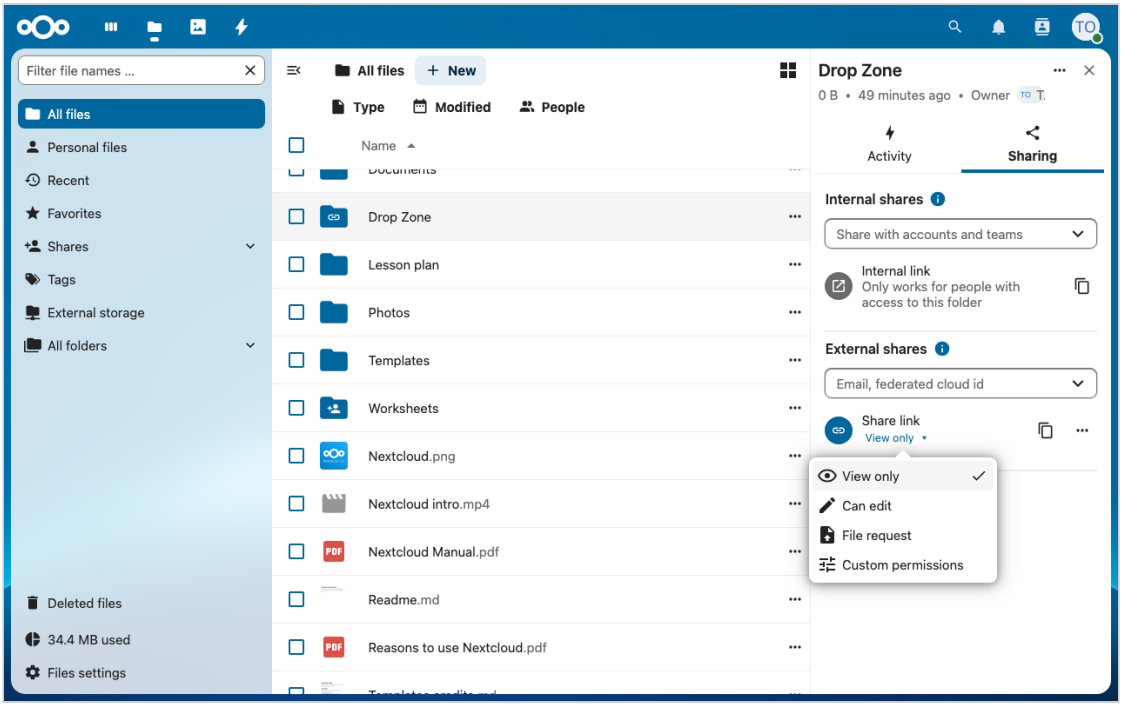


Figure 45. Create link

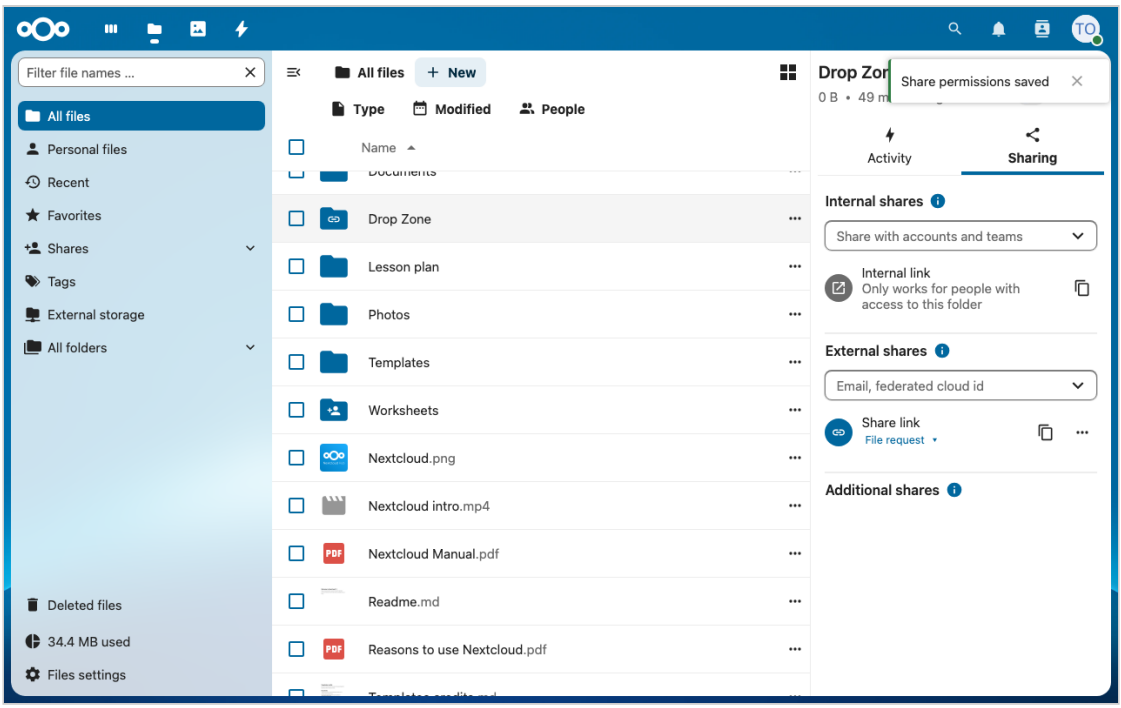


Figure 46. Link created

3. Give learners the link (for example, write it on the board). Uploaded files appear in your `Drop Zone` folder.

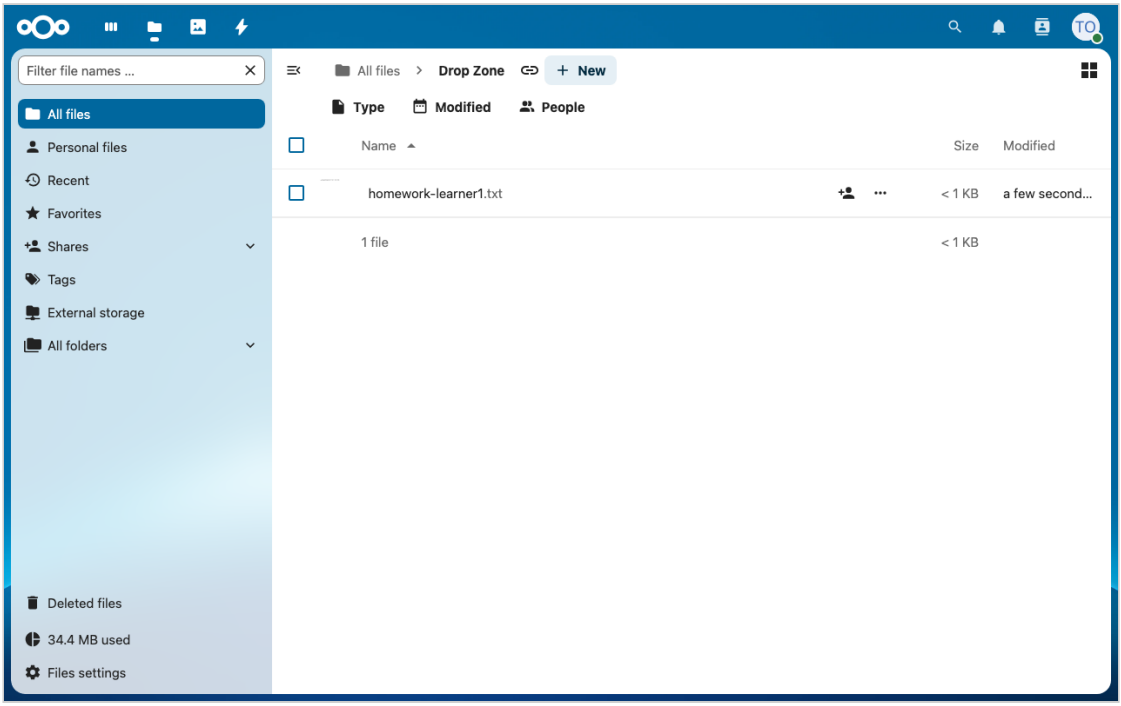


Figure 47. Received homework

6. Nextcloud for learners (try it yourself)

Learners open Nextcloud from the Console, like Kolibri: **IDEA** on the desktop, then **Open** on the Nextcloud card (2.2.1). They have no personal Nextcloud app on the classroom computer, so they always work in the browser.

6.1 Open class files and co-edit

1. Sign in to **Nextcloud Demo** as `learner1`, password `learner1911!` (private window). Shared folders and files appear in **Files** with the teacher's name.

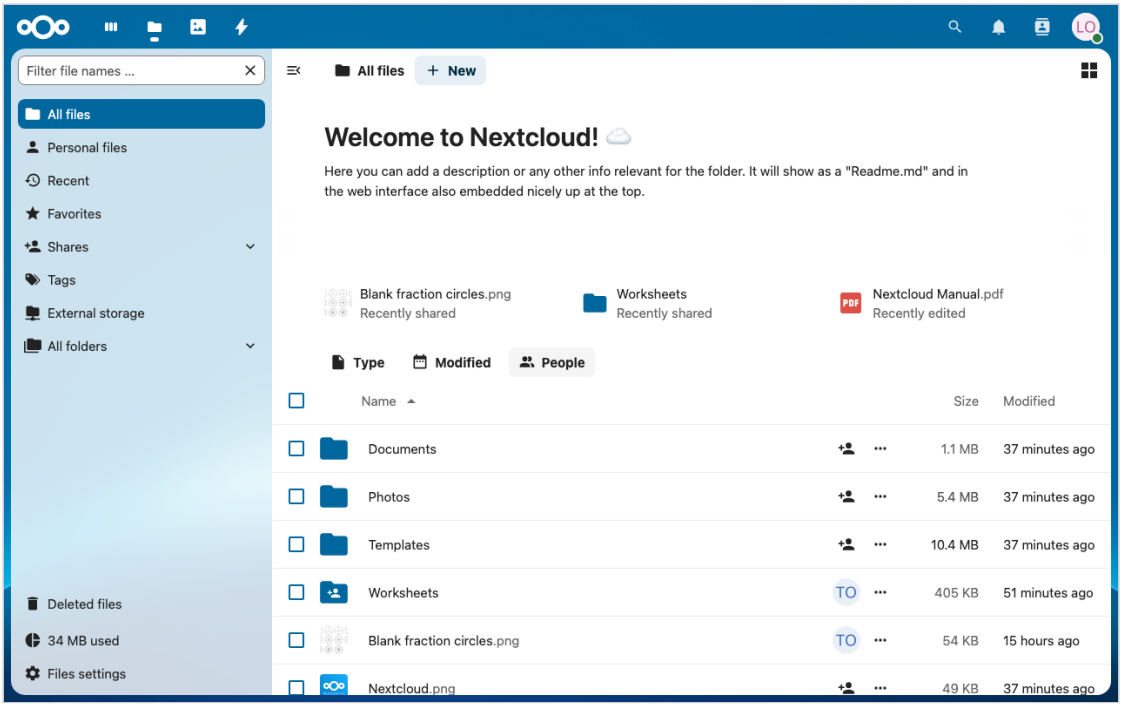


Figure 48. Learner files

2. Open `Worksheets`: the learner can open and download the worksheets, not change them.

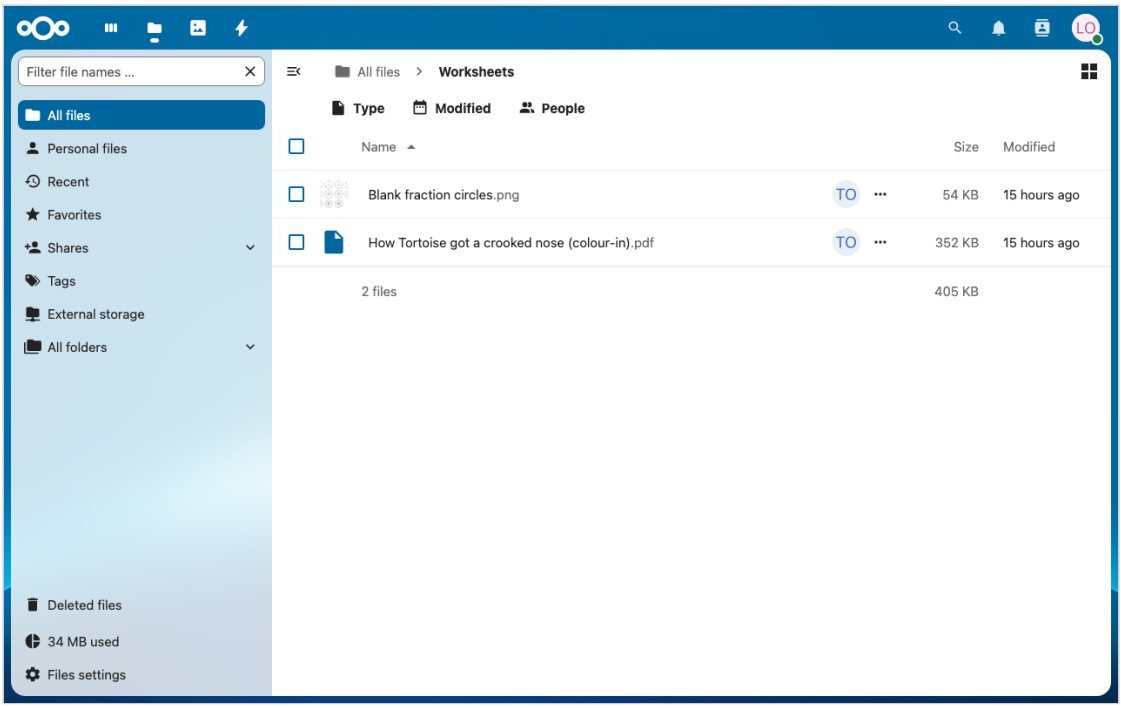


Figure 49. Worksheets

3. Open `Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

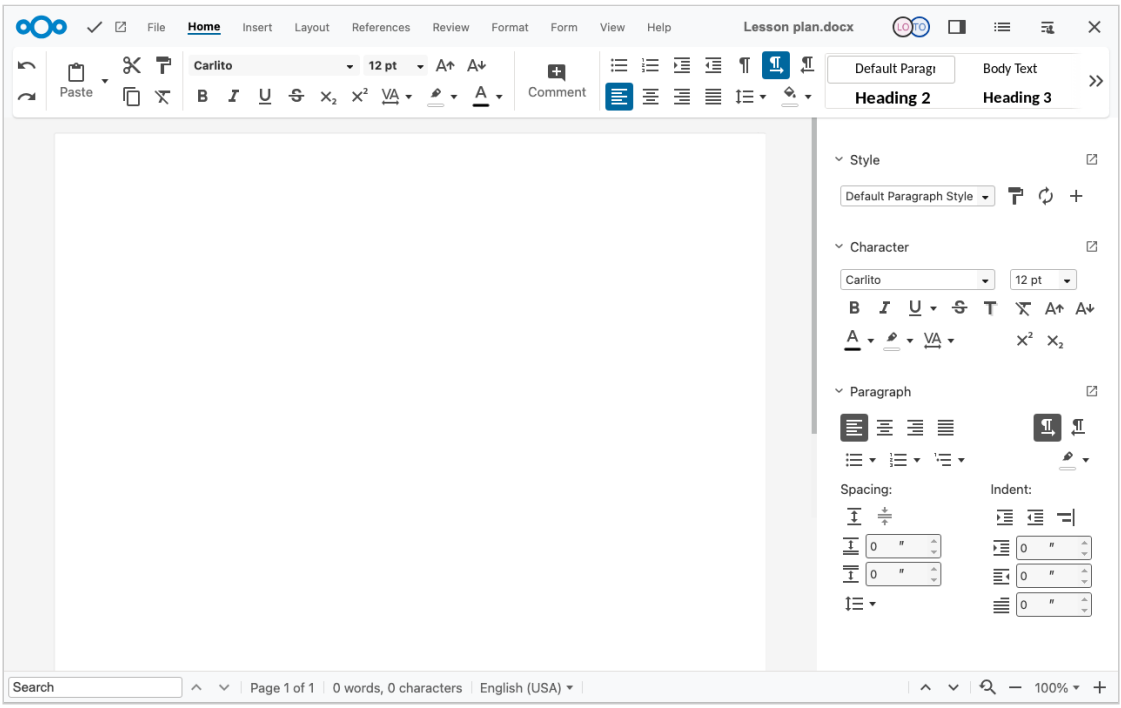


Figure 50. Learner in document

6.2 Hand in homework

1. Open the File drop link from 6.3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

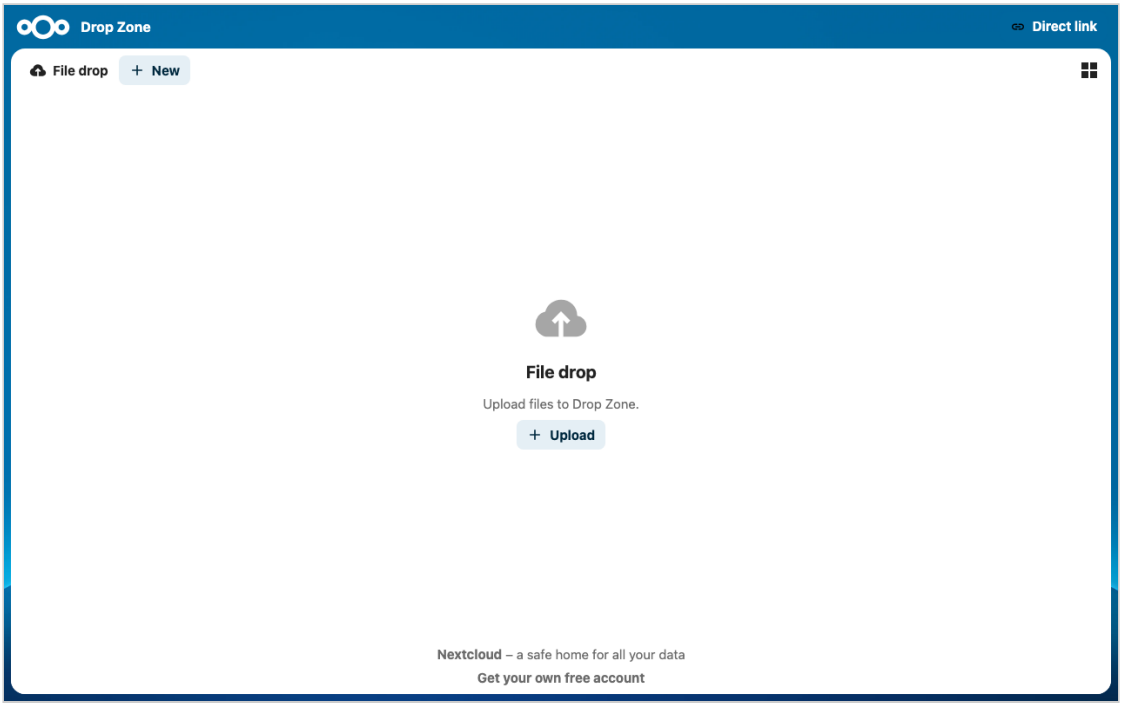


Figure 51. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

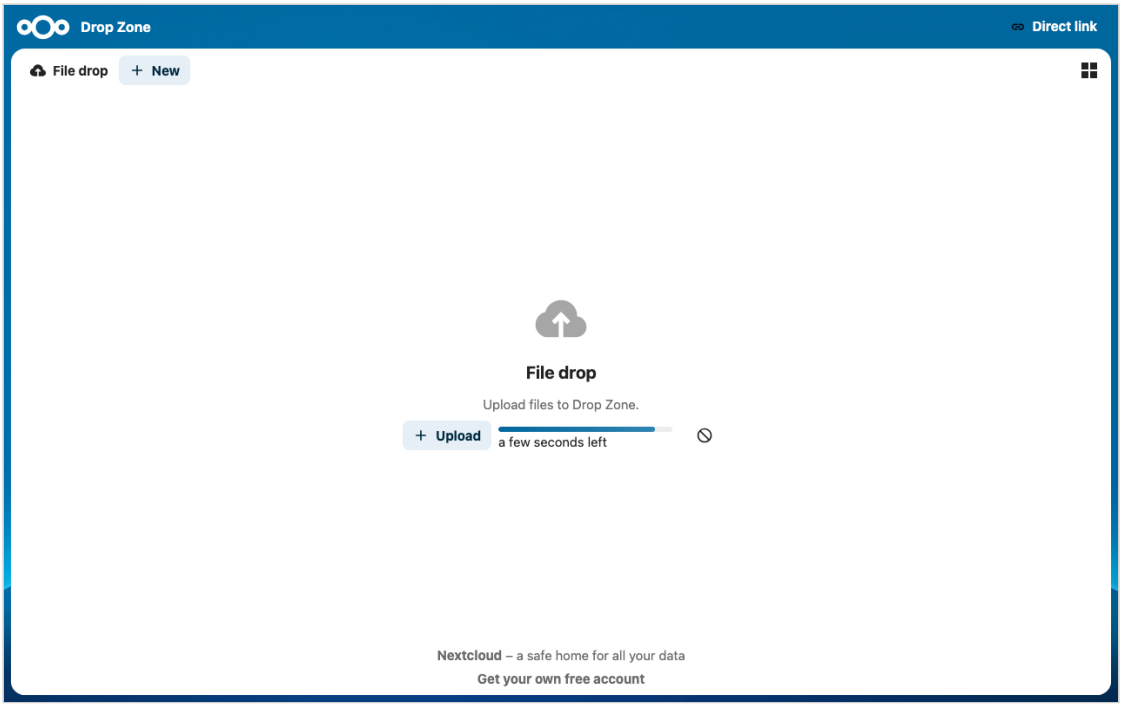


Figure 52. Uploading: the bar shows the progress. When it is gone, the file is in the teacher's folder

The teacher sees the file in `Drop Zone` (5.3).

7. Wikipedia offline

This chapter shows you how to look up an article in Wikipedia and use it in your lesson. Everything works without internet.

The steps below use the **Wikipedia (Demo)** app, which holds **Wikipedia in simple English**: shorter articles in easier English. Your school's own **Wikipedia** app has the full English Wikipedia and works the same way.

7.1 Look something up and use it in class

1. Open **Wikipedia (Demo)** from the Console (chapter 2). The library page shows one book, **Wikipedia in simple English**.

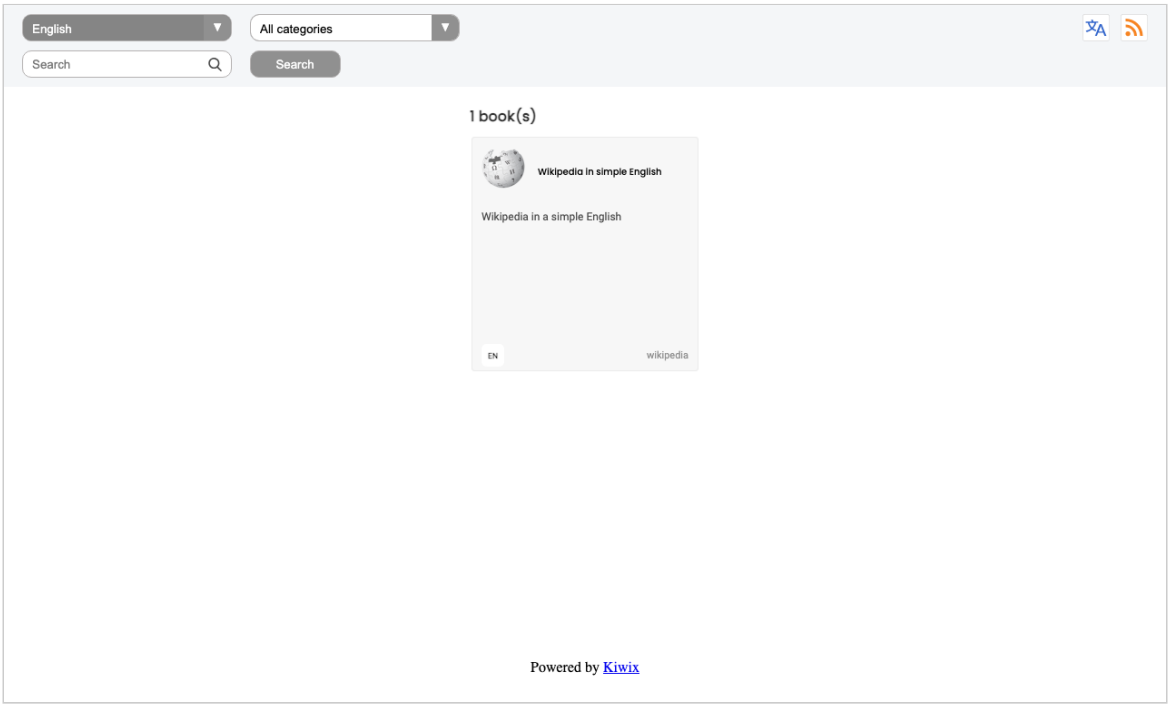


Figure 53. The library page

2. Click **Wikipedia in simple English**. The **Welcome to Wikipedia** page opens, with topics grouped by subject.



Figure 54. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`, and press Enter. In the results, click **Fraction**.

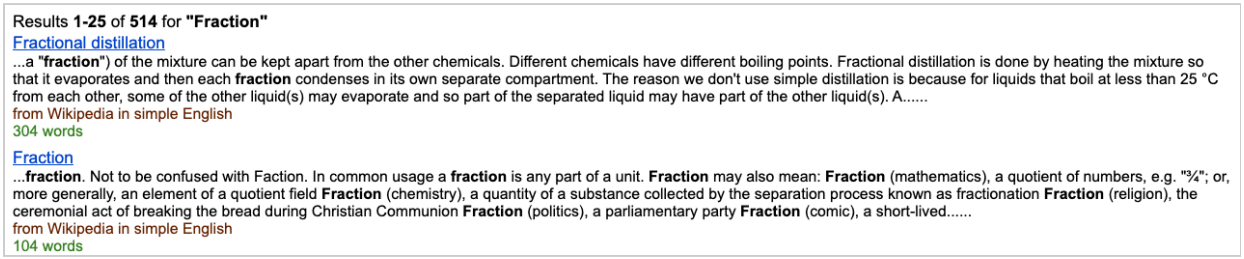


Figure 55. Search results

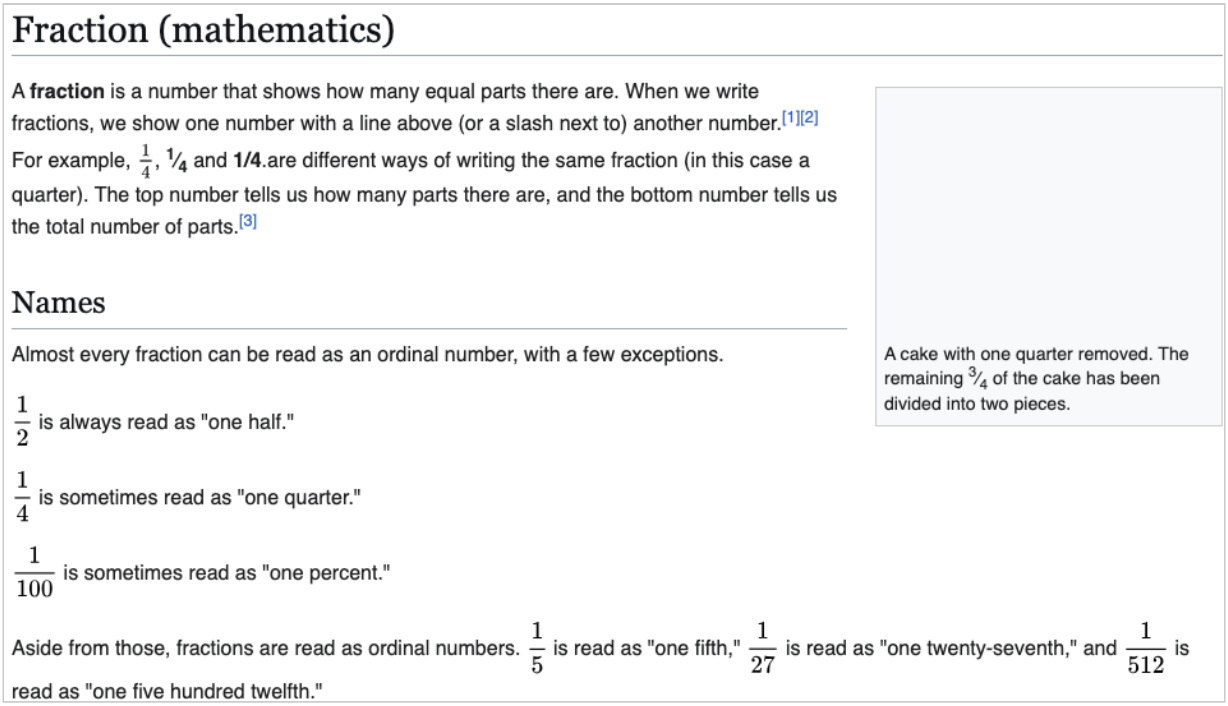
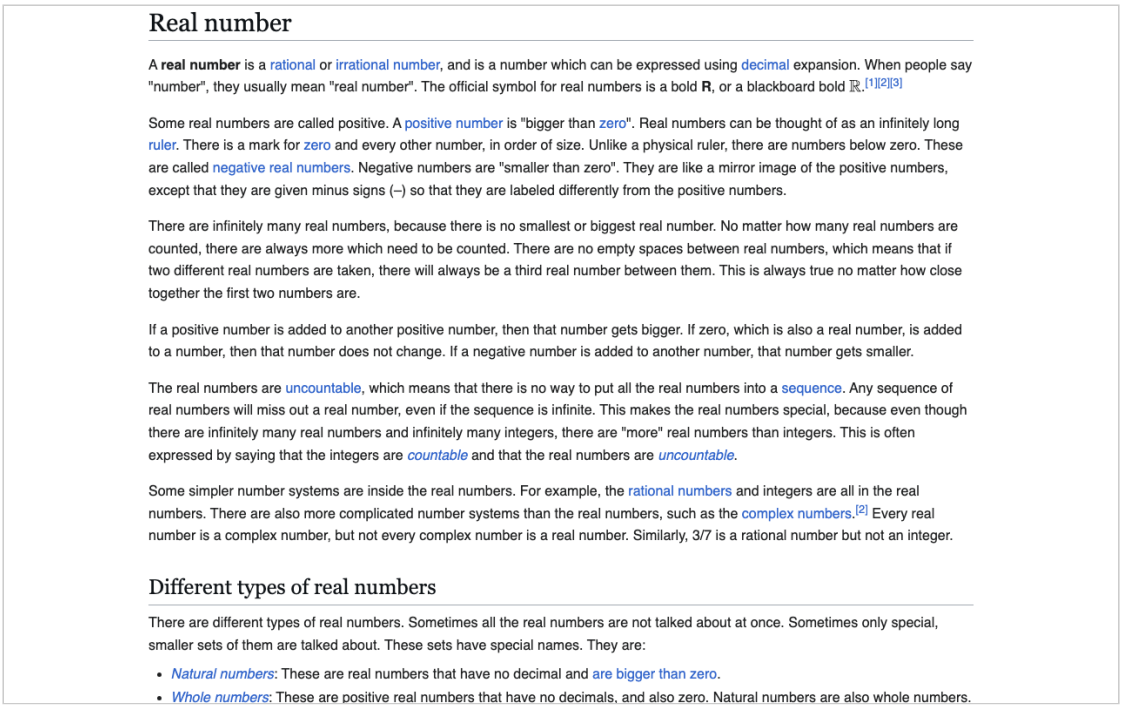


Figure 56. Fraction article

4. Follow blue links to related articles, e.g. **Real number**.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 57. Linked article

> **Classroom idea.** Give learners 3 questions to answer from one article; they read on their own device. If an article is long, point learners to one section.

8. Your own files: the Files Disk

A **Files Disk** is an **IDEA Disk** that the operator prepares in the Console (IDEA for Operators: Tutorial). Its content appears in Nextcloud as the folder **School Files**. Use it to bring in material you downloaded elsewhere, for example worksheets you copied at another school. On the kit, every IDEA box has its own Files Disk.

Good to know:

- **School Files** is only in **Nextcloud**, not in Kolibri.
- Only teachers see the **School Files** folder. Learners get only the files you share with them.
- **The Backup Disk does not save School Files.** Keep your own copy of important files.

8.1 Open School Files

1. Open Nextcloud from the Console and sign in. In **Files** you see the folder **School Files** (marked as **External storage** in the left menu).

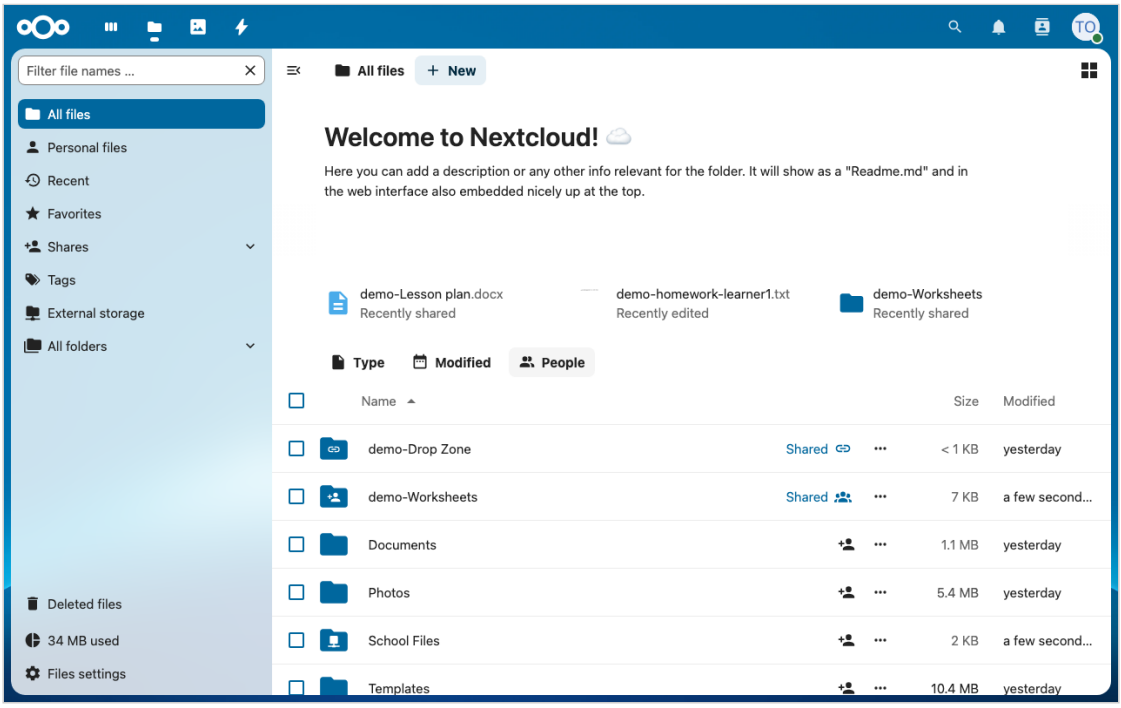


Figure 58. School Files in Nextcloud

2. Open **School Files** to see what is on the disk (here `Fractions-Worksheet-1.pdf`).

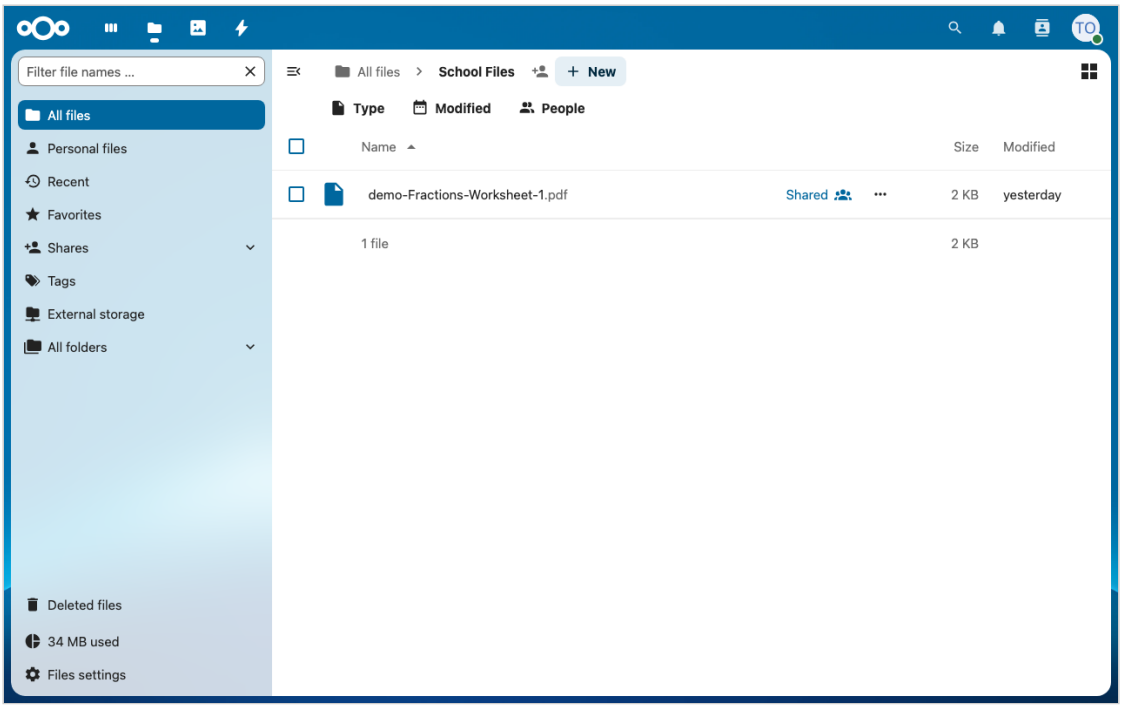


Figure 59. School Files content

8.2 Share a file from School Files with your class

1. On the file's row click **··· (Actions) > Show sharing options**, or click the share icon.
2. In **Internal shares**, type the class group name (here `Grade 5`; in Nextcloud Demo on the kit, `learners`) and choose it from the list. Keep **View only**. The group then appears under **Internal shares**.

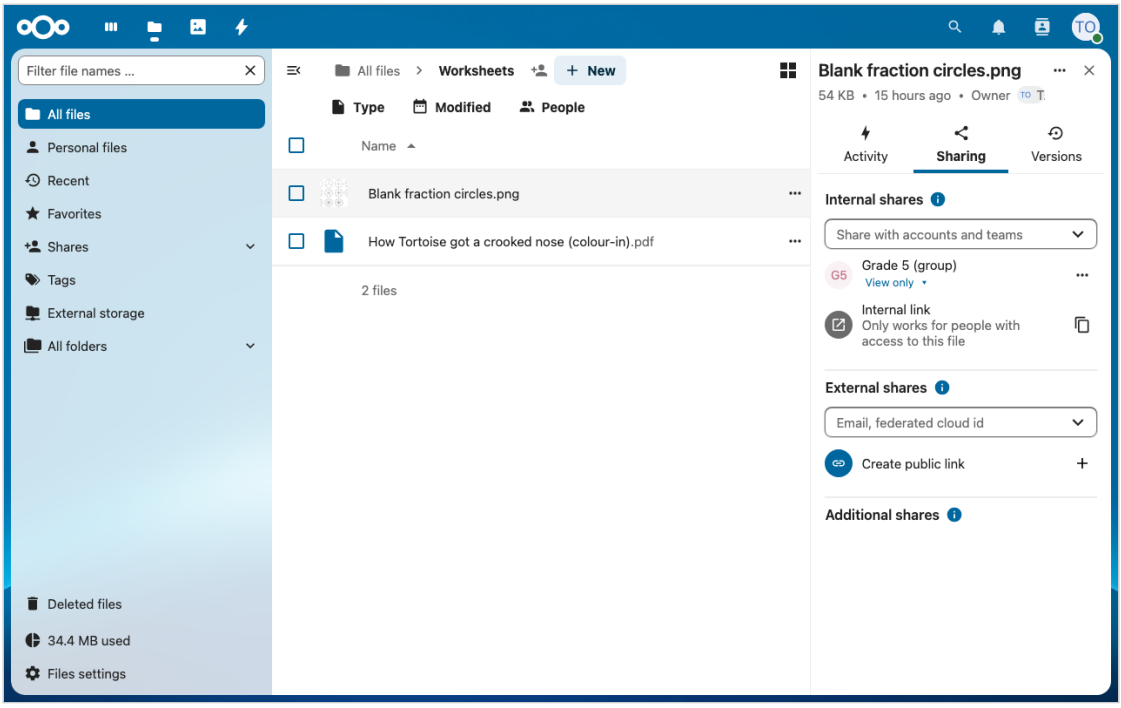


Figure 60. A file shared with Grade 5 (here a worksheet)

3. A learner in that group signs in and finds the file in **All files** and under **Recently shared**.

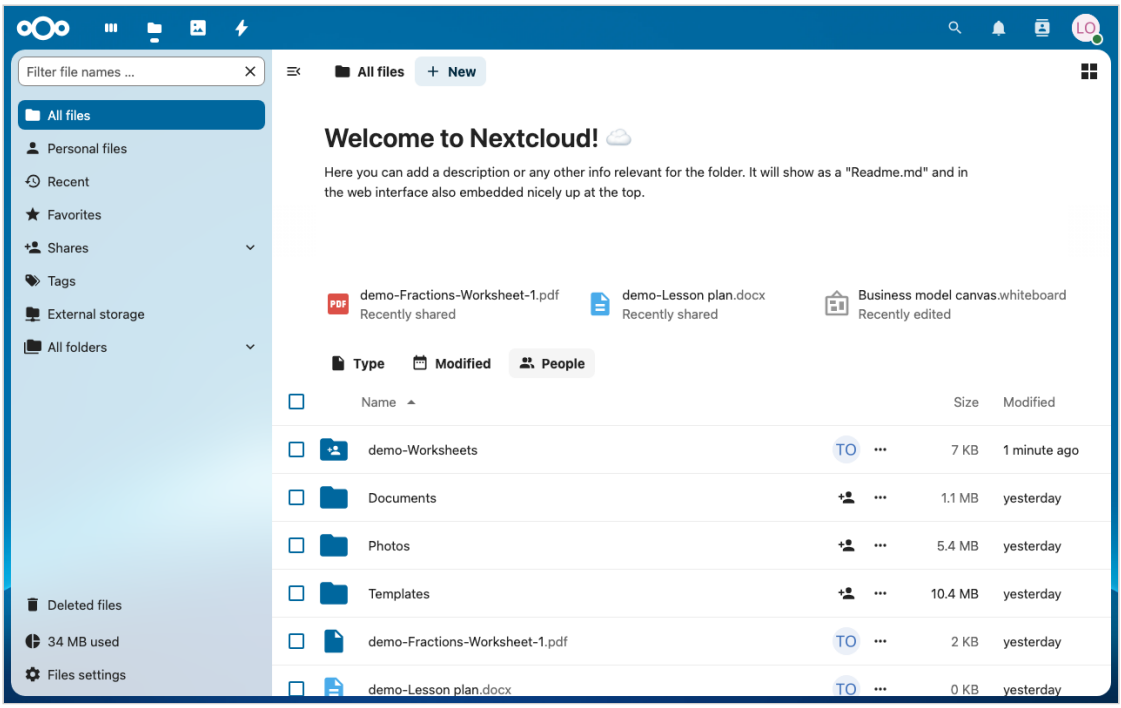


Figure 61. A learner sees the shared file

9. Keeping data safe: the Backup Disk

The operator looks after backups (IDEA for Operators: Tutorial). As a teacher you should know:

- A **Backup Disk** is a separate **IDEA Disk** with the **backup** badge in the Console.
- It saves **every app in full**: Kolibri (classes, learners, progress, and the channels), Nextcloud (accounts, and the files inside Nextcloud) and Wikipedia. On the kit that is about 164 GB per IDEA box, which fits on its 240 GB Backup Disk.
- It does **not** save the **School Files** folder (chapter 8).
- **During its backup, an app is stopped and can't be opened.** It starts again by itself when its backup is done. The first backup of all apps takes about 2 h 15 min, so backups are planned outside lesson time.
- How often backups run (on request, each time the disk is plugged in, or on a schedule) is set by your IDEA contact.
- If the IDEA box breaks, the operator restores the latest backup onto a spare disk. Only the work done after the last backup is lost.
- Never unplug a disk while a backup is running.

Appendix: demo accounts

This appendix lists the demo accounts and the demo content. The demo setup is on every IDEA box: the apps **Kolibri Demo** and **Nextcloud Demo**. Use them to follow the tutorial.

The demo school in Kolibri Demo is **IDEA Demo School**. The same usernames work in Kolibri Demo and in Nextcloud Demo.

Passwords:

- In Kolibri, learners have no password. They sign in with only their username.
- Every other demo password is the account name followed by `911!`, for example `teacher1911!`.

Username	Password	App	Role	Used in
admin	admin911!	Kolibri Demo	Kolibri admin of IDEA Demo School	3.2, 3.3, 4
teacher1	teacher1911!	Kolibri Demo	Coach of class Grade 5	3
teacher2	teacher2911!	Kolibri Demo	Coach of class Grade 5	3
learner1	none (username only)	Kolibri Demo	Learner in Grade 5	3.5, 4
learner2	none (username only)	Kolibri Demo	Learner in Grade 5	4
learner3	none (username only)	Kolibri Demo	Learner in Grade 5	3.5, 3.6
learner4	none (username only)	Kolibri Demo	Learner in the school, not in Grade 5	3.3
admin	admin911!	Nextcloud Demo	Nextcloud administrator	
teacher1	teacher1911!	Nextcloud Demo	Teacher (group teachers)	5, 8
teacher2	teacher2911!	Nextcloud Demo	Teacher (group teachers)	5.2
learner1	learner1911!	Nextcloud Demo	Learner (group learners)	6, 8.2
learner2	learner2911!	Nextcloud Demo	Learner (group learners)	6.1
learner3	learner3911!	Nextcloud Demo	Learner (group learners)	

Demo content:

- Kolibri: the school **IDEA Demo School** with the class **Grade 5**.

- Kolibri Demo channels: **Maths**, **Grade 5 Science** and **Early Reading**, and also **Khan Academy (English - US curriculum)** in the Library. **Maths** has **Fractions** (three pictures, three short videos and **Fractions practice**), **Multiplication** (one video) and **Times Tables** (two documents and **Times tables practice**). **Early Reading** has **Our Country** (two picture activities) and **Stories** (seven story documents and the exercise **Reading: Rudo and the goat**). The goal for **Fractions practice** and **Times tables practice** is **Get 4 correct**.
- Kolibri Demo lessons for Grade 5: **Maths lesson** (the 11 Maths resources), **Grade 5 Science lesson** and **Early Reading lesson** (10 items). The **Maths lesson** and the quiz are already visible to learners. You can make a smaller **Fractions lesson** yourself in chapter 3; this tutorial uses the ready-made **Maths lesson**.
- Kolibri Demo quiz: **Grade 5 Practice Quiz** (10 questions: 4 from Rudo and the goat, 6 from Fractions practice), already started for Grade 5. `learner3` has submitted it (7 out of 10) and has completed **Fractions practice** and **Cake in quarters**, so reports show results.
- `learner4` (Learner Four) is in the school but not in Grade 5, so chapter 3.3 can show the enroll step.
- Nextcloud Demo: the folders **Worksheets** and **Teacher Notes** for `teacher1` and `teacher2`. You make the **Drop Zone** folder and the document **Lesson plan** yourself in chapter 5.
- Wikipedia (Demo): Wikipedia in simple English.

The school's own apps (not demo) start almost empty: **Kolibri** has only `admin` (the school creates its teachers and learners), **Nextcloud** has `admin`, `teacher1` and `learner1`. The passwords follow the same rule. Learners need a password in Nextcloud, not in Kolibri.

The pictures in the demo content are from Zimbabwe for now. The file `CREDITS.txt` in the demo content gives the source and licence of every file.